

Digital IC Design Portfolio

Full-custom CMOS logic, sequential circuits, and a 256-bit SRAM designed and verified in Cadence Virtuoso

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Project Overview

This portfolio documents a progression from transistor-level CMOS cells to an integrated static random-access memory. I designed and verified reusable combinational and sequential circuits, then integrated a 6T bitcell with row decoding, precharge, write, and sensing circuitry to form a 16×16 SRAM array.

Category	Details
Scope	Schematic capture, custom layout, circuit simulation, DRC, and LVS
Designs	Inverter, NAND2, XOR, transmission gate, D flip-flop, 6T SRAM cell, decoder, sense amplifier, precharge, write driver, SRAM column, and 16×16 array
Key results	6T SRAM SNM of 210.8107 mV, DFF hold time of 7 ps, and an integrated capacity of 256 bits
Software	Cadence Virtuoso and Cadence ADE

Project Summary

The work combines transistor-level circuit design, reusable symbol creation, custom physical layout, simulation-driven characterization, and hierarchical memory integration. The following pages retain the complete revised technical record, including schematics, layouts, waveforms, and verification results.

Architecture and Design Progression

Stage	Implemented blocks	Verification
CMOS logic	Inverter, NAND2, and XOR	DC or transient simulation, DRC, and LVS
Sequential library	Transmission gate and D flip-flop	Transient simulation, hold-time characterization, DRC, and LVS
SRAM subsystem	6T cell, NAND5, 4-to-16 decoder, BLSA, precharge, write driver, SRAM column, and 16 × 16 array	SNM and read/write simulations, block-level DRC/LVS, and top-level integration

Selected Engineering Outcomes

Reusable cell hierarchy. Primitive CMOS gates were converted into symbols and reused to construct higher-level logic and memory-control circuits.

Timing characterization. The D flip-flop was evaluated around the active clock edge using a parametric input-delay sweep, producing a measured hold time of 7 ps under the documented simulation conditions.

Memory stability and operation. The 6T bitcell produced a simulated static noise margin of 210.8107 mV. Separate simulations demonstrate write, read, precharge, differential sensing, and data-retention behavior.

Hierarchical integration. Sixteen rows and sixteen columns were combined with the peripheral circuits to create a 256-bit SRAM architecture.

Verification Status

DRC and LVS were completed for the individual logic cells and SRAM peripheral blocks documented in the report.

Technical Report Preview

Part I CMOS Logic and Sequential Circuits

Transistor-level design and verification of the inverter, NAND2, XOR, transmission gate, and D flip-flop. This section includes schematic and symbol views, transient simulations, physical layouts, DRC/LVS evidence, and hold-time characterization.

Part II 16 × 16 SRAM

Design and characterization of the 6T SRAM bitcell, 4-to-16 row decoder, bit-line sense amplifier, precharge circuit, write driver, SRAM column, and the integrated 256-bit array. The report includes static-noise-margin analysis, read/write timing, custom layouts, and physical-verification results.

Key Results at a Glance

Result	Value or status
D flip-flop hold time	7 ps
6T SRAM static noise margin	210.8107 mV
Memory organization	16 rows × 16 columns
Total capacity	256 bits
Block-level verification	DRC and LVS completed for the documented logic and peripheral blocks
Top-level verification	Read/write simulation and DRC review completed; LVS remained unresolved at the project deadline

Document Organization

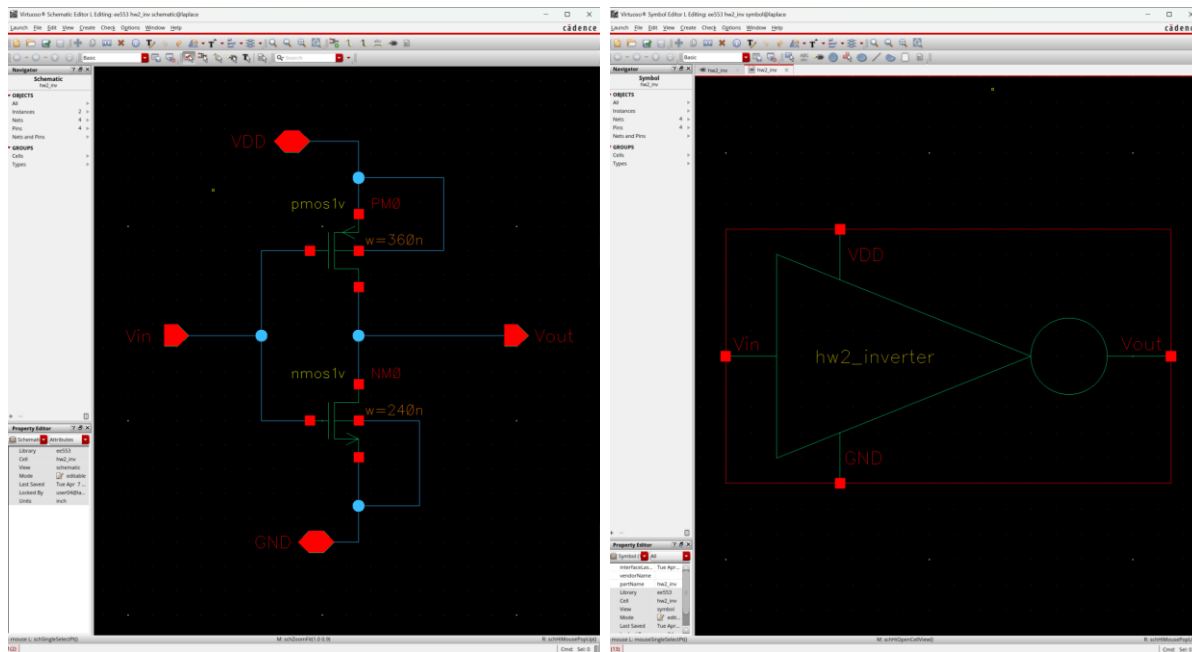
The revised technical report begins on the next page. Figures are presented in the same order as the submitted design record so that each schematic, simulation, layout, and verification result can be reviewed together.

Digital IC Design Portfolio

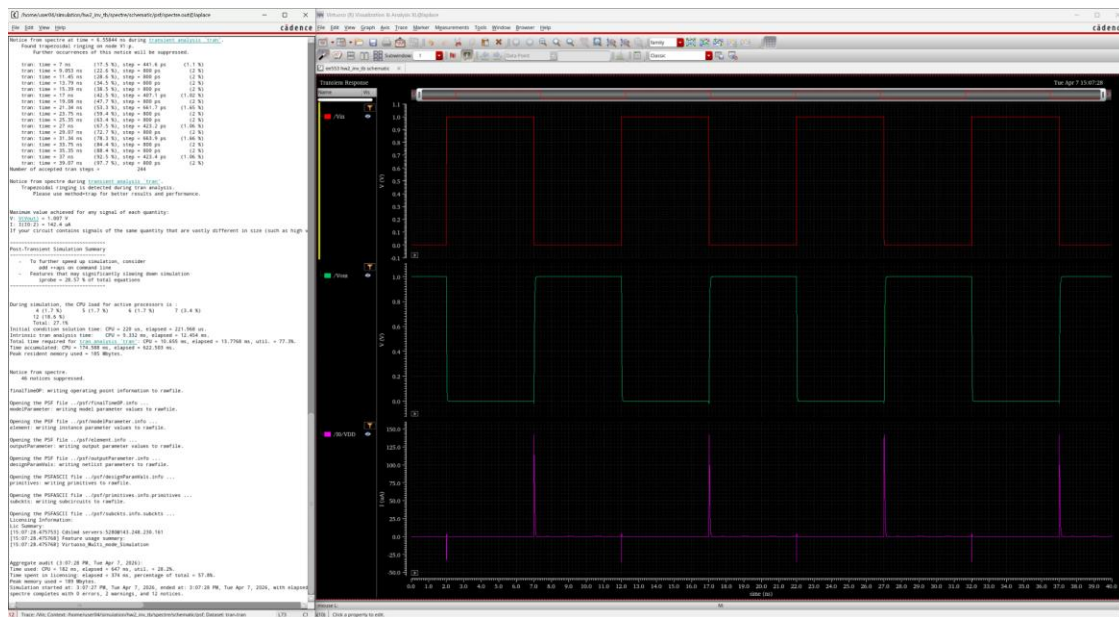
Full-custom CMOS logic, sequential circuits, and a 256-bit SRAM designed and verified in Cadence Virtuoso

Inverter

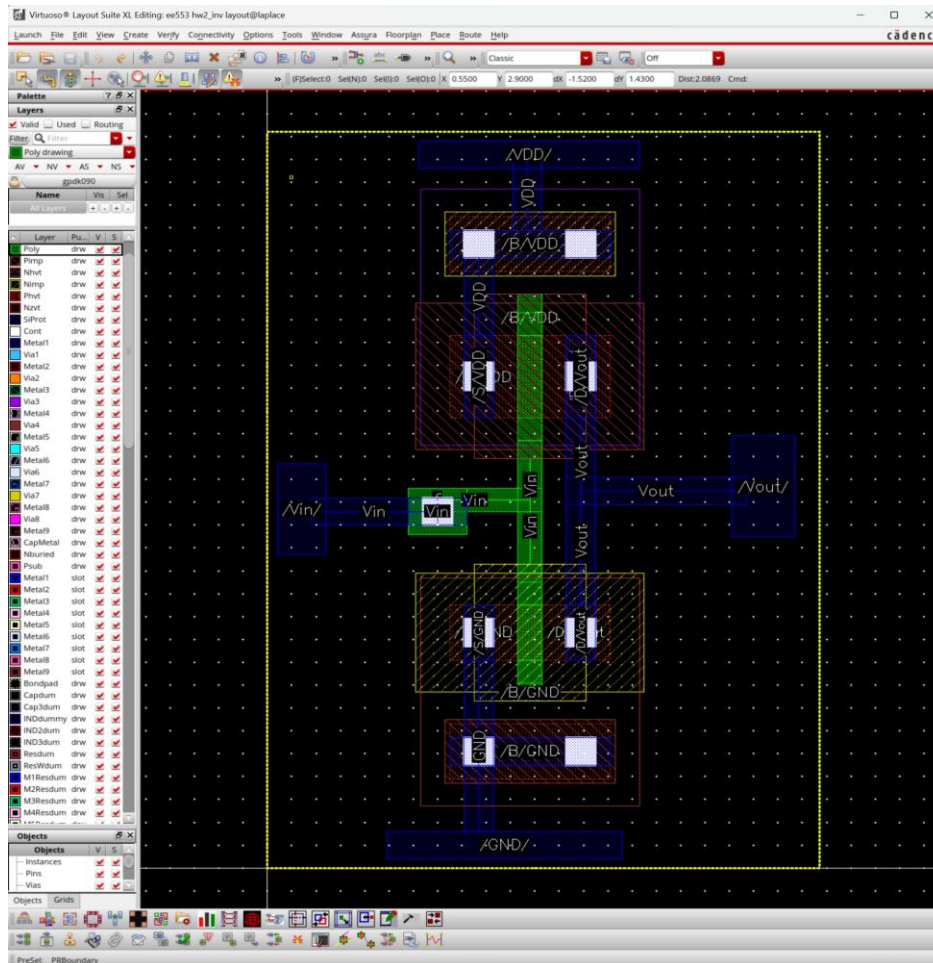
Schematic and Symbol



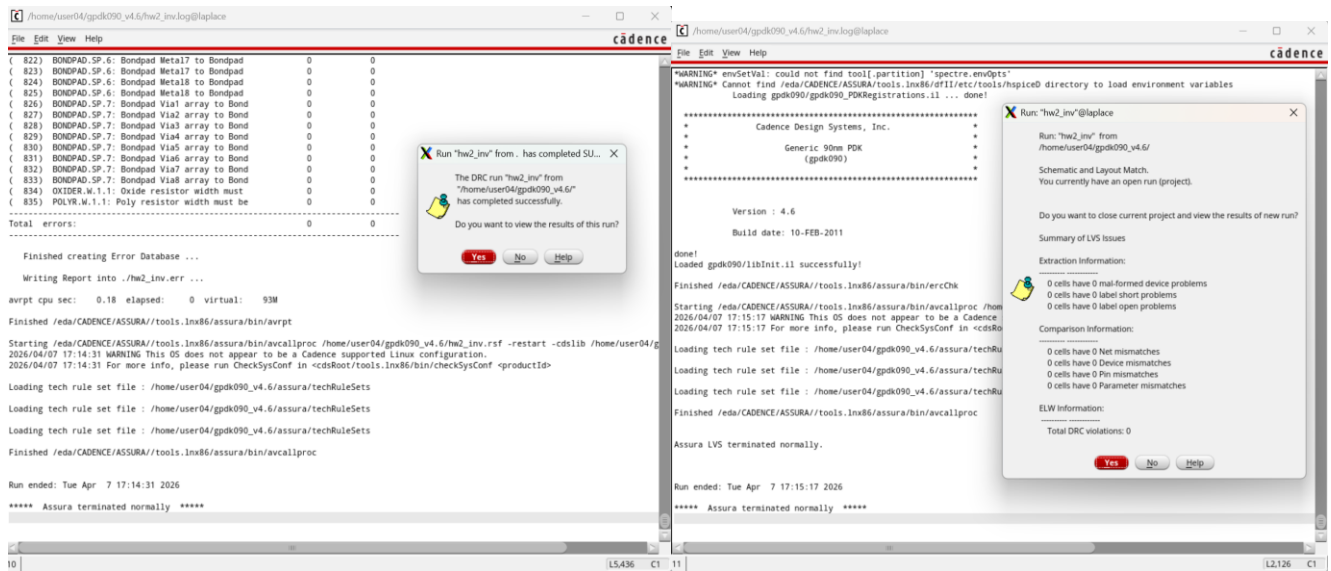
Transient Simulation



Layout

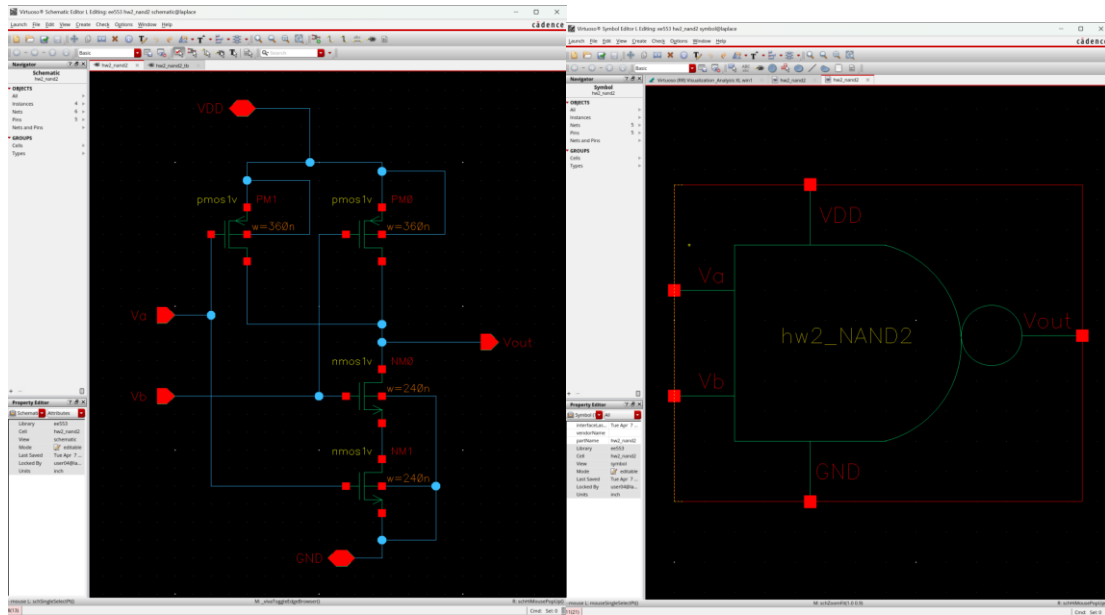


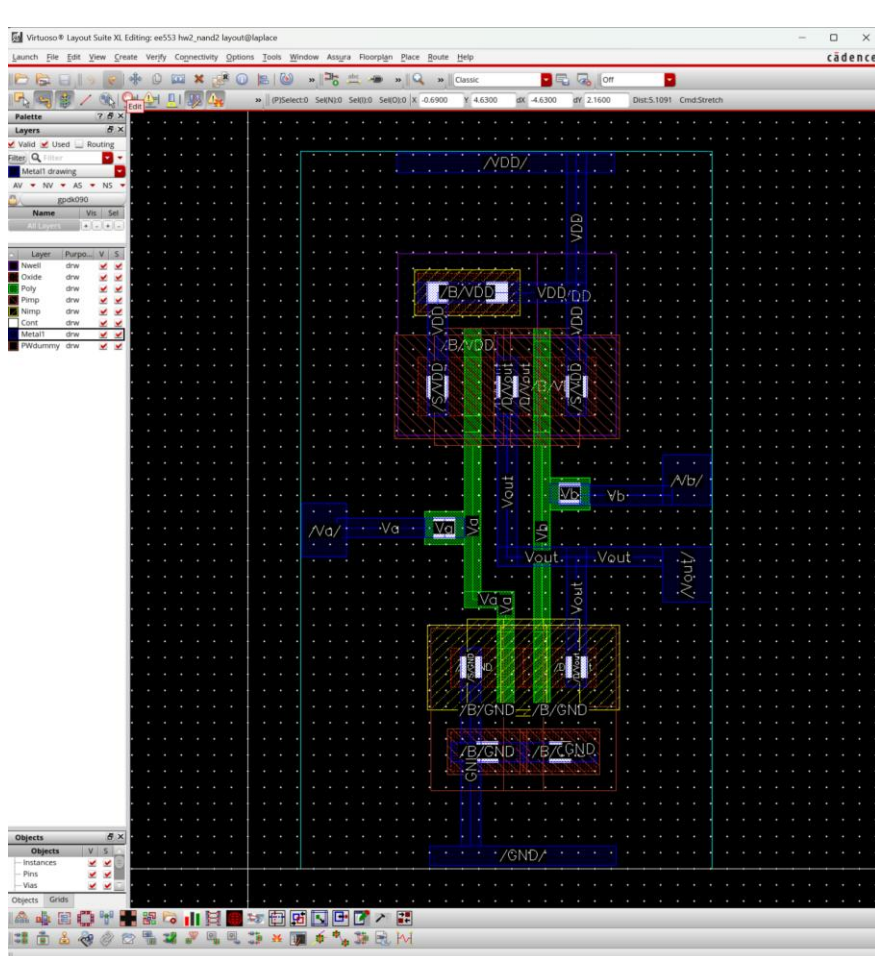
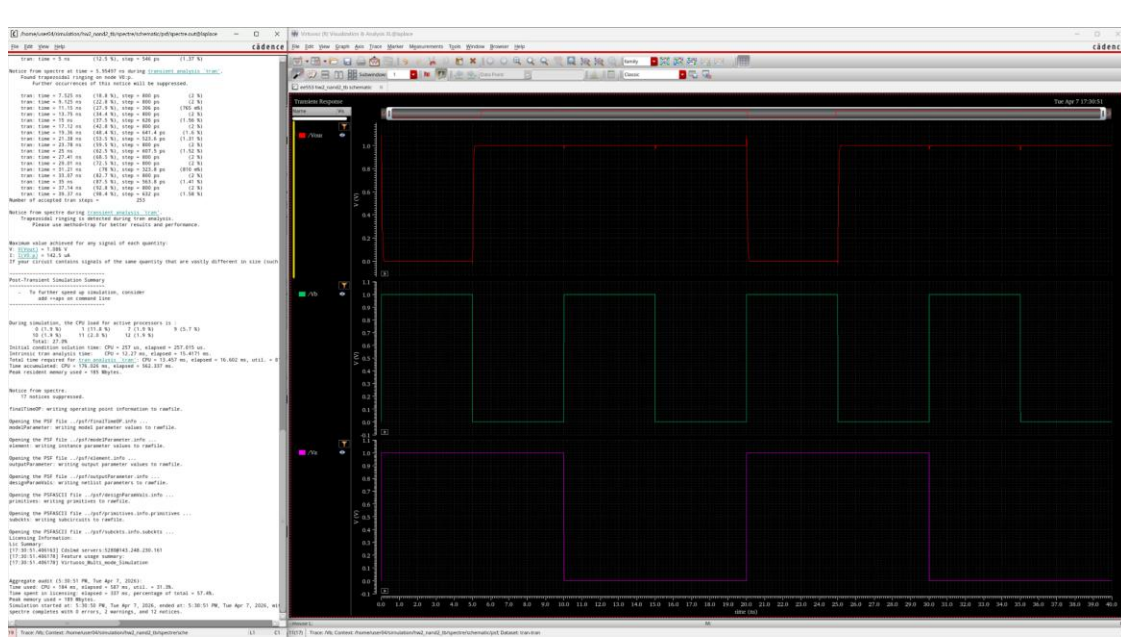
DRC and LVS Results



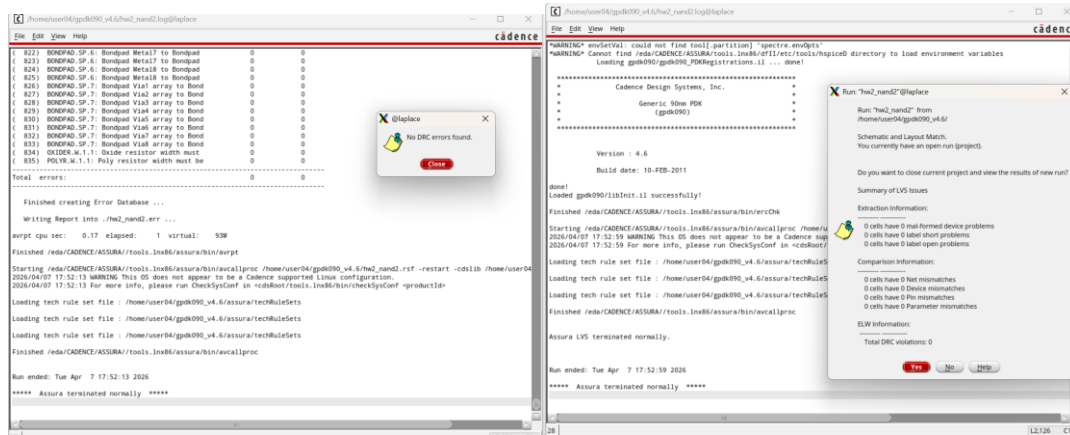
NAND 2

Schematic and Symbol



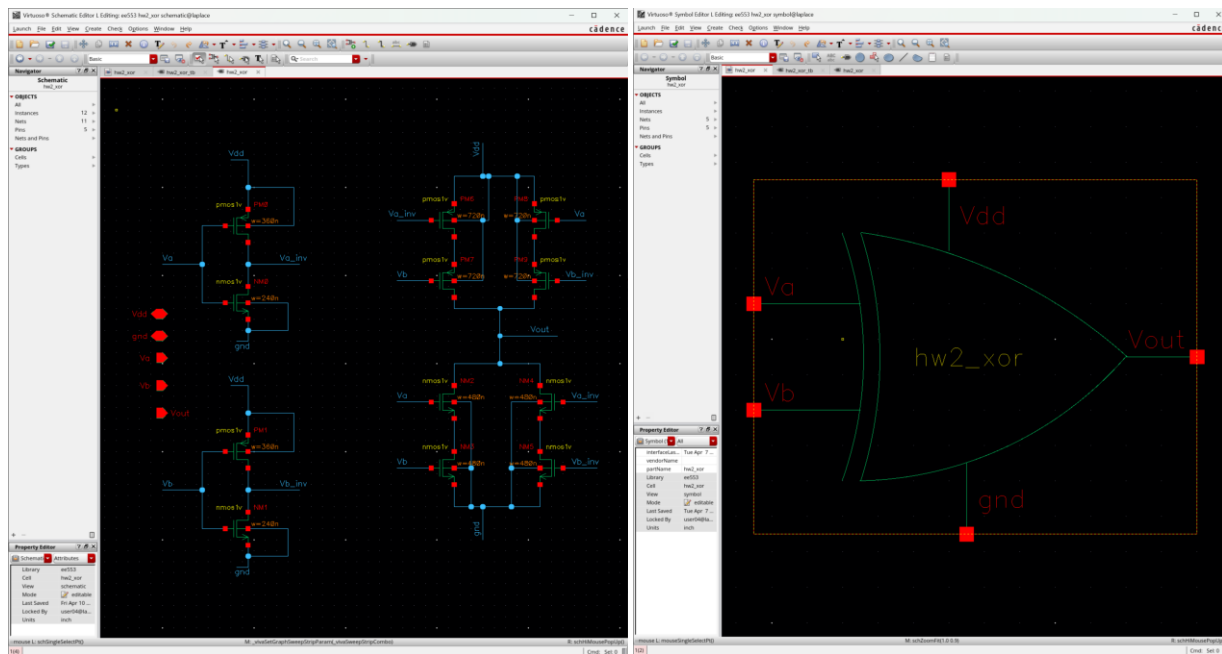


DRC and LVS Results

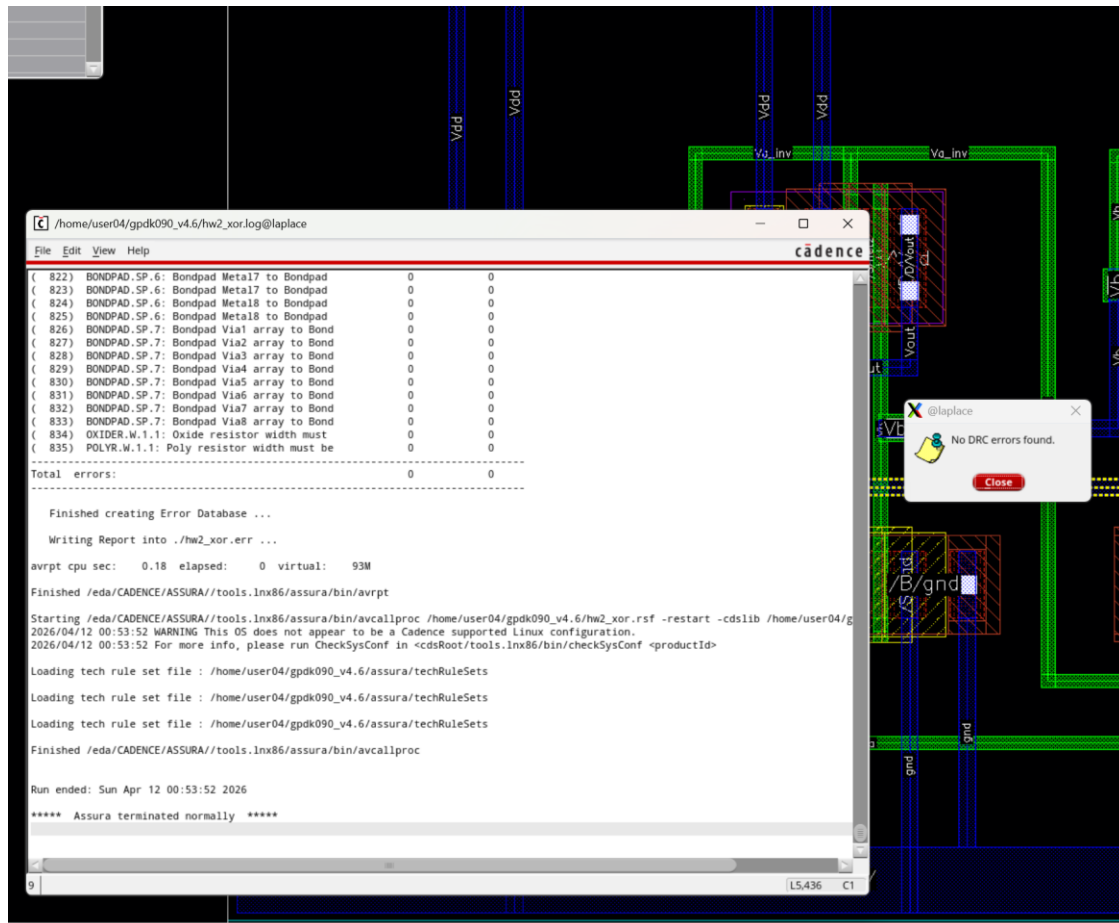
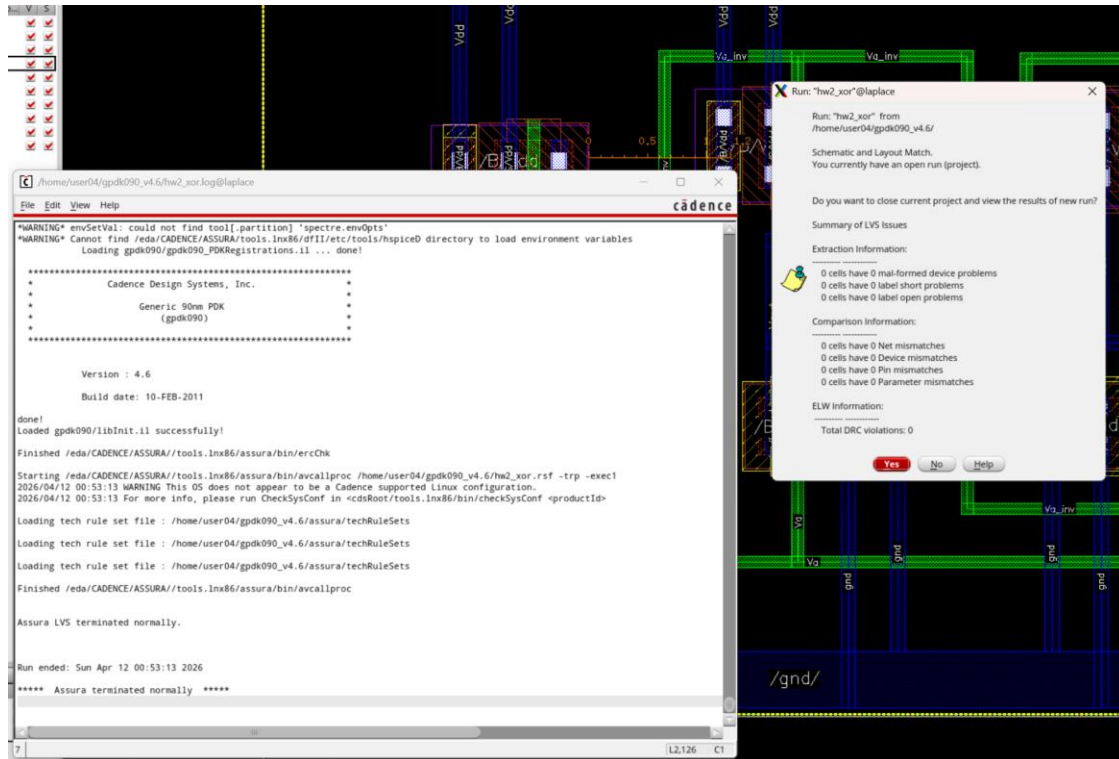


2-input XOR

Schematic and Symbol

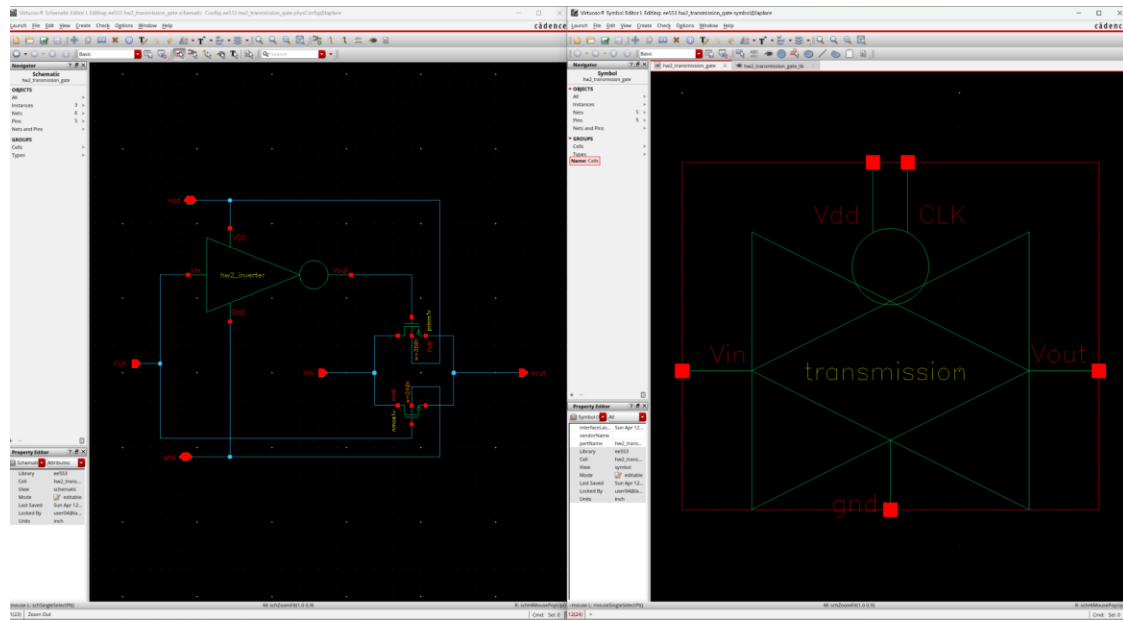


DRC and LVS Results

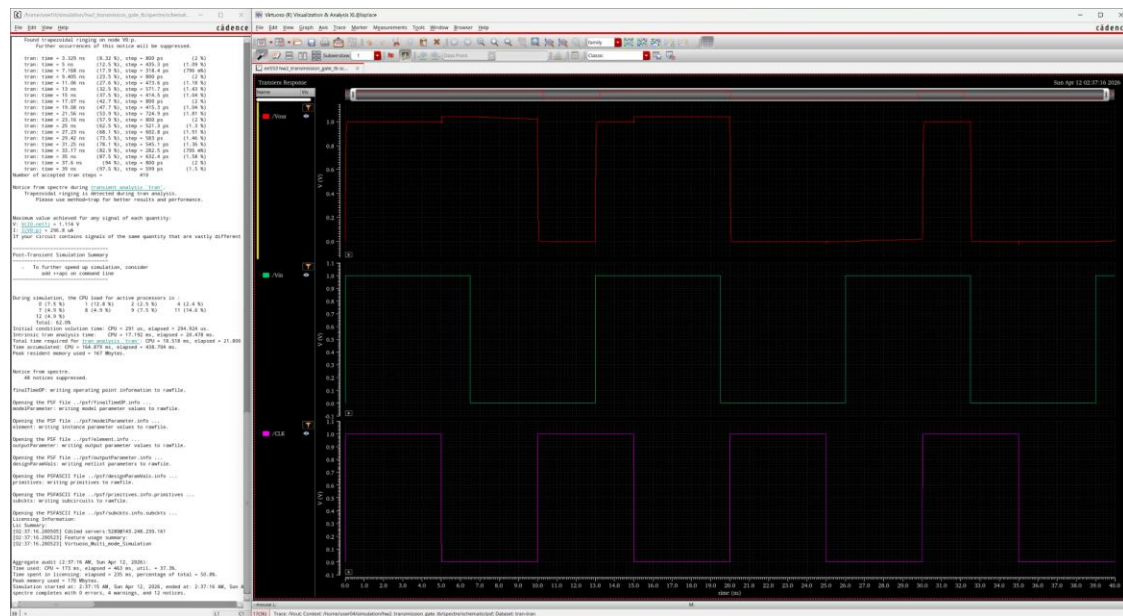


Transmission Gate

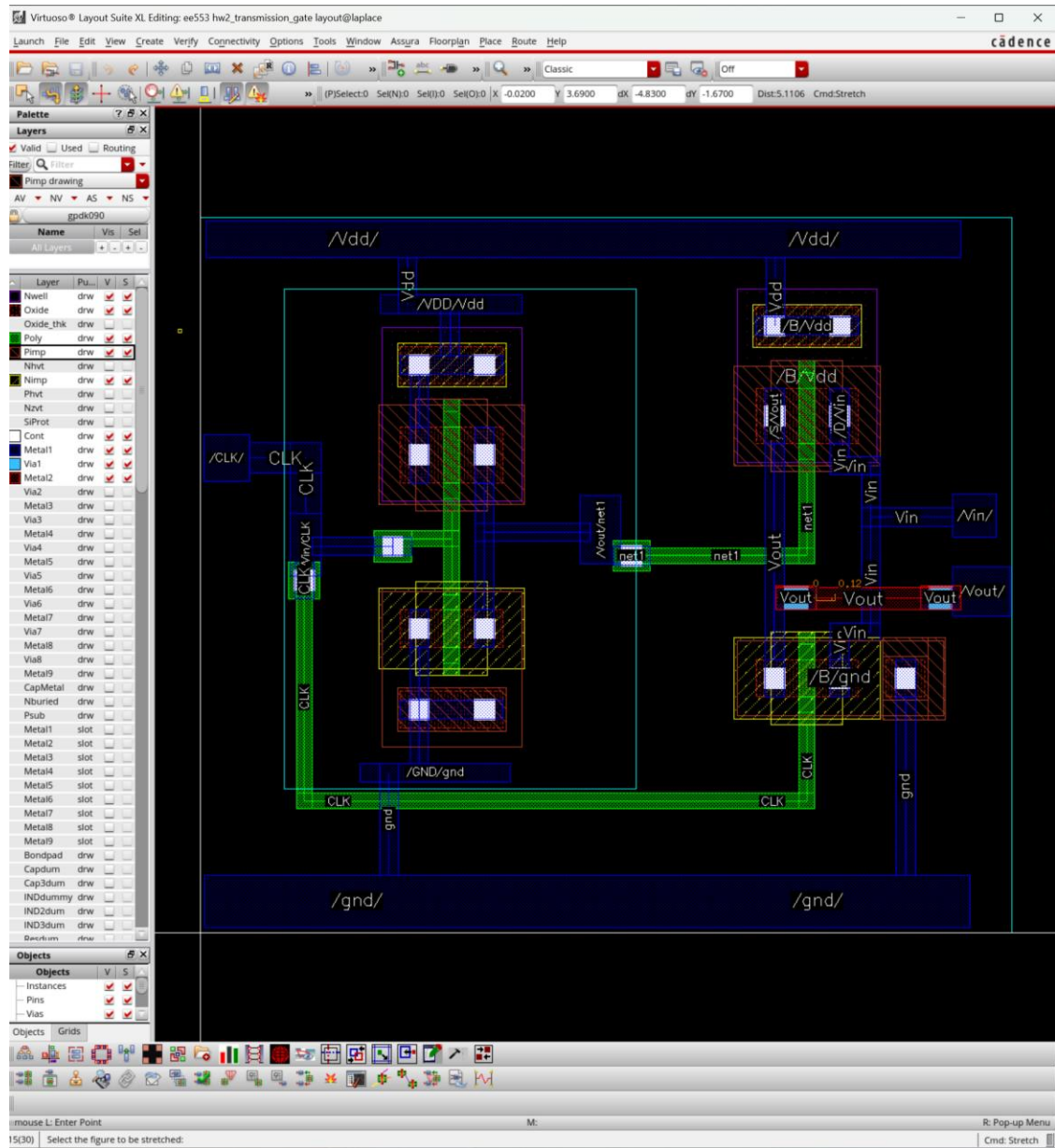
Schematic and Symbol



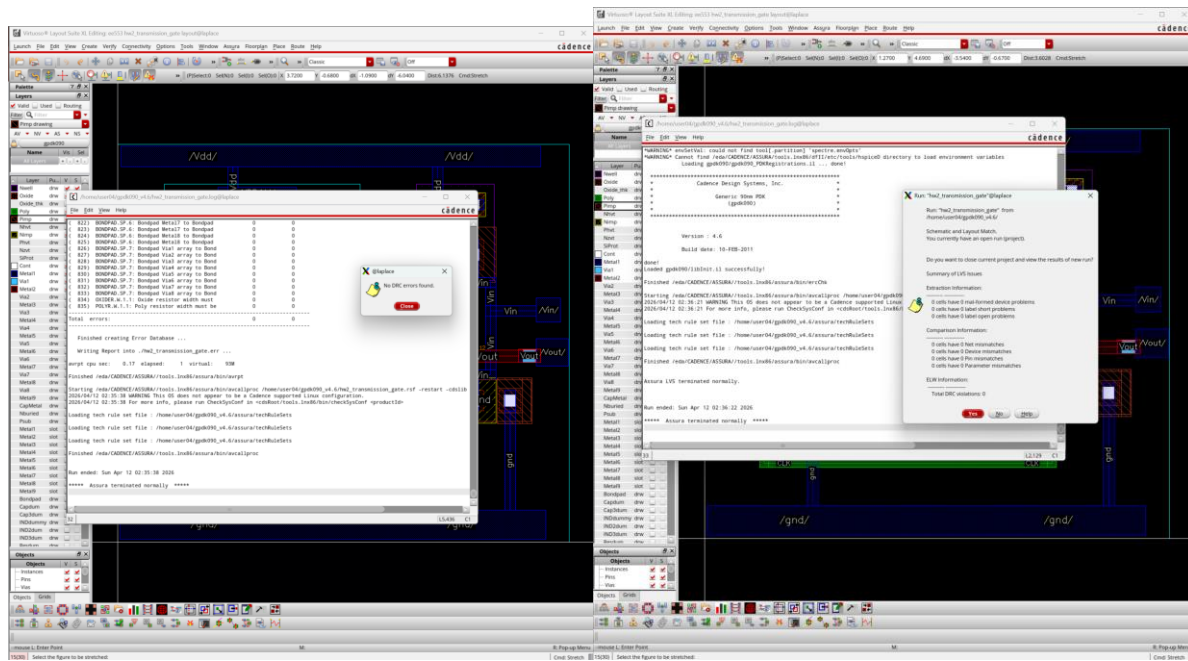
Transient Response



Layout

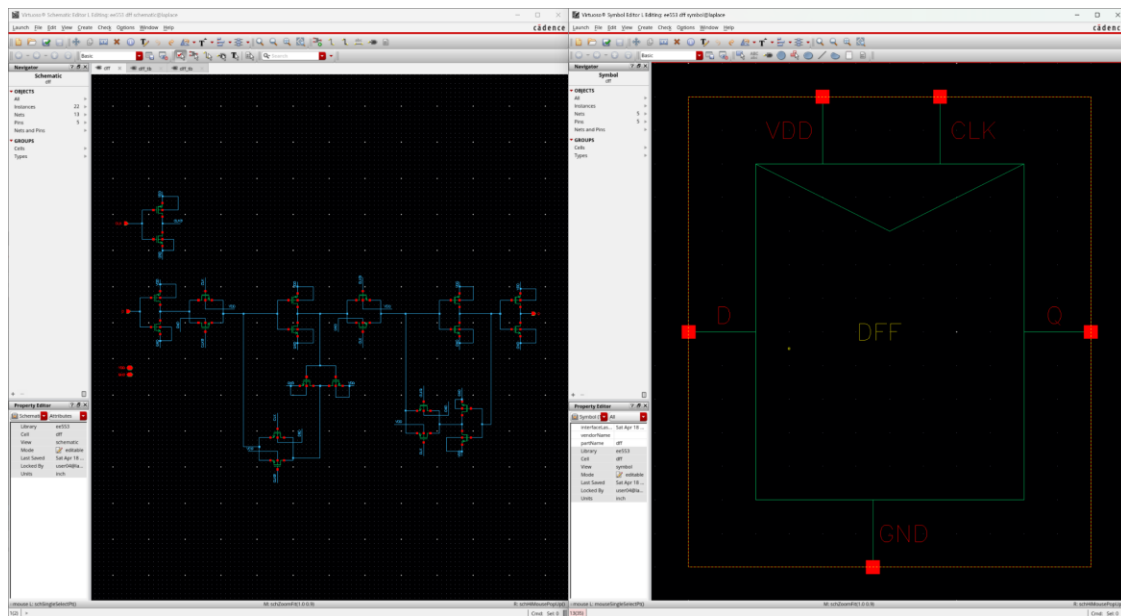


DRC and LVS results

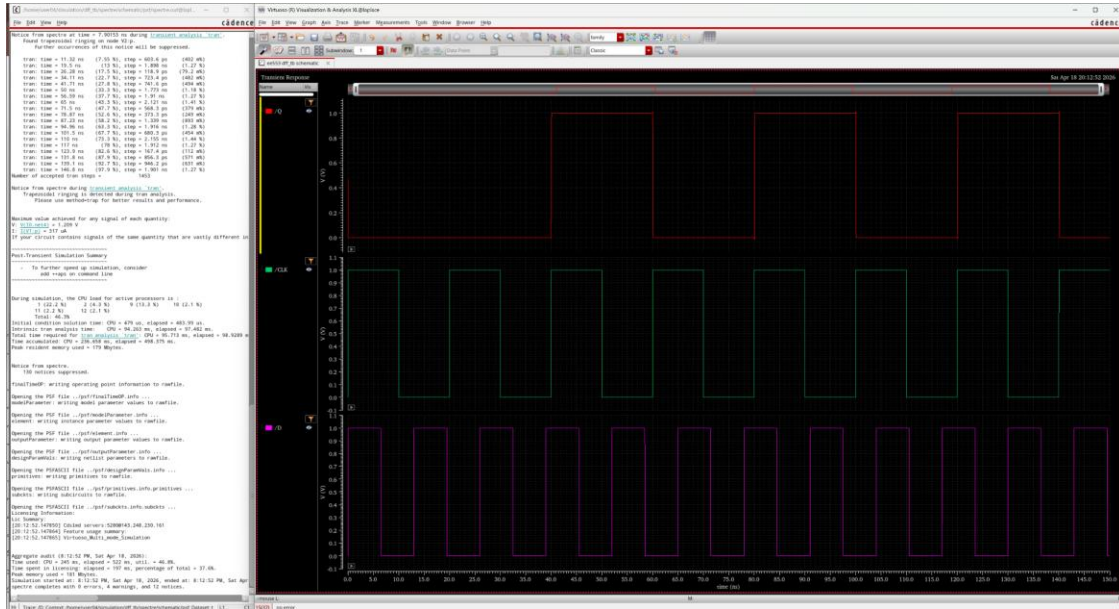


D-Flip Flop

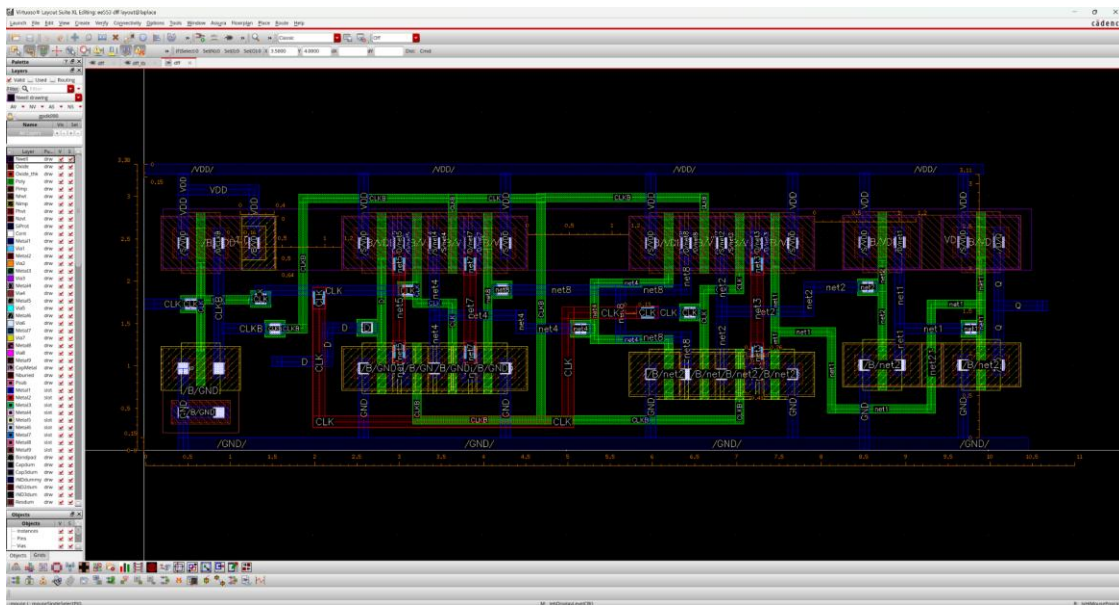
Schematic & Symbol



Transient Response



Layout



DRC and LVS Results

```
/home/user04/gpd090_v4.6/dff.log@laplace
File Edit View Help cadence

( 822) BONDPAD.SP.6: Bondpad Metal17 to Bondpad      0      0
( 823) BONDPAD.SP.6: Bondpad Metal17 to Bondpad      0      0
( 824) BONDPAD.SP.6: Bondpad Metal18 to Bondpad      0      0
( 825) BONDPAD.SP.6: Bondpad Metal18 to Bondpad      0      0
( 826) BONDPAD.SP.7: Bondpad Via1 array to Bond      0      0
( 827) BONDPAD.SP.7: Bondpad Via2 array to Bond      0      0
( 828) BONDPAD.SP.7: Bondpad Via3 array to Bond      0      0
( 829) BONDPAD.SP.7: Bondpad Via4 array to Bond      0      0
( 830) BONDPAD.SP.7: Bondpad Via5 array to Bond      0      0
( 831) BONDPAD.SP.7: Bondpad Via6 array to Bond      0      0
( 832) BONDPAD.SP.7: Bondpad Via7 array to Bond      0      0
( 833) BONDPAD.SP.7: Bondpad Via8 array to Bond      0      0
( 834) OXIDER.W.1.1: Oxide resistor width must      0      0
( 835) POLYR.W.1.1: Poly resistor width must be     0      0
-----
Total errors:                                     0      0
-----

Finished creating Error Database ...

Writing Report into ./dff.err ...

avrpt cpu sec:   0.18 elapsed:   0 virtual:   93M

Finished /eda/CADENCE/ASSURA//tools.Inx86/assura/bin/avrpt

Starting /eda/CADENCE/ASSURA//tools.Inx86/assura/bin/avcallproc /home/user04/gpd090_v4.6/dff.rsf -restart -cdslib /home/user04/gpd0
2026/04/18 20:09:21 WARNING This OS does not appear to be a Cadence supported Linux configuration.
2026/04/18 20:09:21 For more info, please run CheckSysConf in <cdsRoot/tools.Inx86/bin/checkSysConf <productId>

Loading tech rule set file : /home/user04/gpd090_v4.6/assura/techRuleSets

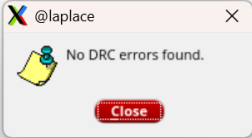
Loading tech rule set file : /home/user04/gpd090_v4.6/assura/techRuleSets

Loading tech rule set file : /home/user04/gpd090_v4.6/assura/techRuleSets

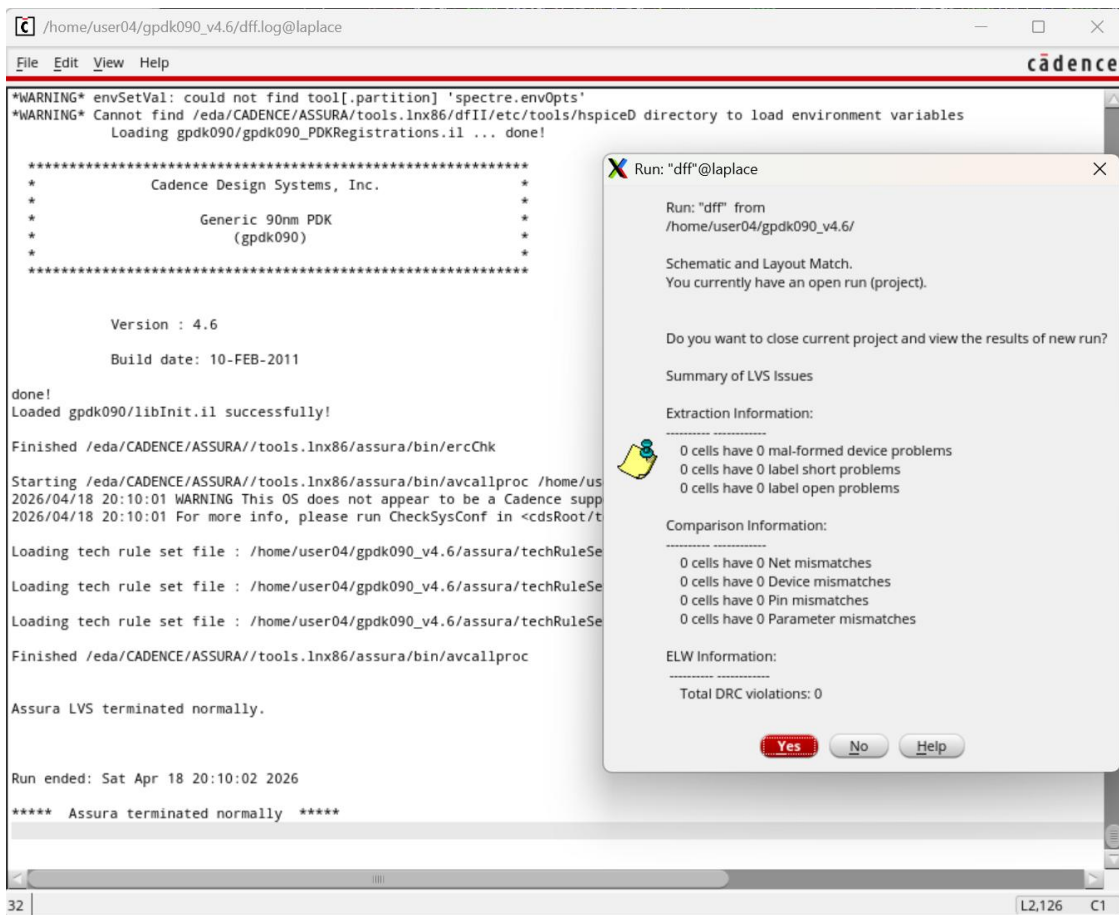
Finished /eda/CADENCE/ASSURA//tools.Inx86/assura/bin/avcallproc

Run ended: Sat Apr 18 20:09:21 2026

***** Assura terminated normally *****
```



```
31 | L5,436 C1
```



Hold time measurement

First, we had to find the nominal CLK-to-Q delay. The Clock and input D characteristics were set, respectively, as follows:

The image shows two side-by-side windows titled "Edit Object Properties@laplace". Each window contains configuration options for a component from the "analogLib" library, specifically the "vpulse" cell. The left window is for instance "V1" and the right for instance "V0".

Common Settings (Both Windows):

- Apply To:** only current (selected), instance (selected)
- Show:** system (unchecked), user (checked), CDF (checked)
- Property Table:**

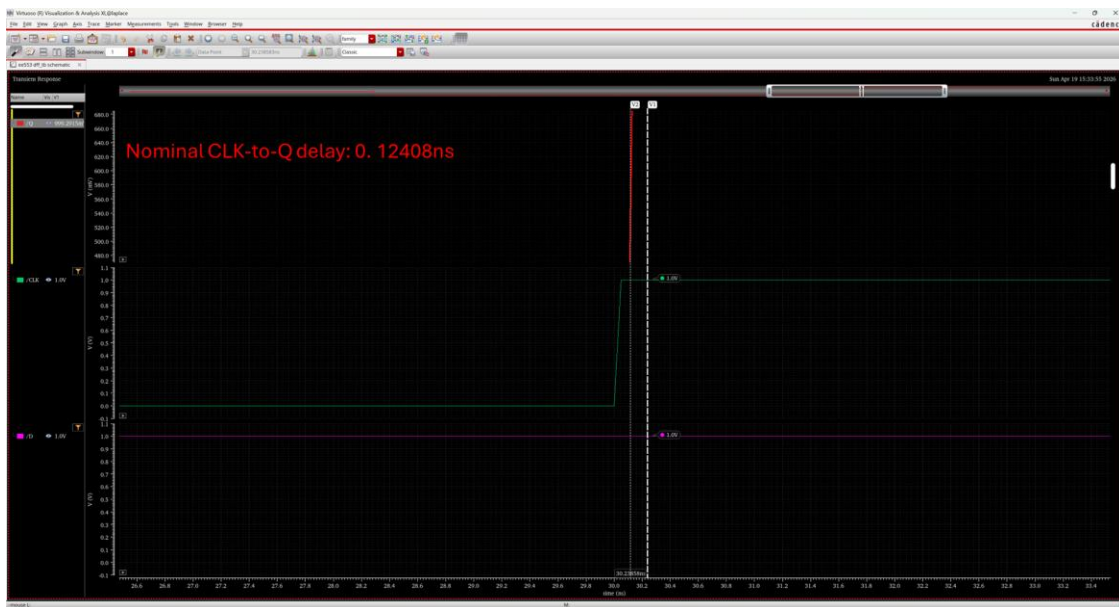
Property	Value	Display
Library Name	analogLib	off
Cell Name	vpulse	off
View Name	symbol	off
Instance Name	V1 (left) / V0 (right)	off
- User Property:**

User Property	Master Value	Local Value	Display
Isignore	TRUE		off

CDF Parameters (Both Windows):

CDF Parameter	Value	Display
Frequency name for 1/period		off
Noise file name		off
Number of noise/freq pairs	0	off
DC voltage		off
AC magnitude		off
AC phase		off
XF magnitude		off
PAC magnitude		off
PAC phase		off
Voltage 1	0 V	off
Voltage 2	1 V	off
Period	20n s	off
Delay time	10n s	off
Rise time	50p s	off
Fall time	50p s	off
Pulse width		off
Temperature coefficient 1		off
Temperature coefficient 2		off
Nominal temperature		off
Type of rising & falling edge		off

At the bottom of each window are buttons: OK, Cancel, Apply, Defaults, Previous, Next, Help.



Then, the input D parameters were set as follows:

Edit Object Properties@laplace

Apply To

only current

instance

Show

☐ system
 ☒ user
 ☒ CDF

Browse

Reset Instance Labels Display

Property	Value	Display
Library Name	analogLib	off
Cell Name	vpulse	off
View Name	symbol	off
Instance Name	V0	off

Add

Delete

Modify

User Property	Master Value	Local Value	Display
lvignore	TRUE		off

CDF Parameter	Value	Display
Frequency name for 1/period		off
Noise file name		off
Number of noise/freq pairs	0	off
DC voltage		off
AC magnitude		off
AC phase		off
XF magnitude		off
PAC magnitude		off
PAC phase		off
Voltage 1	0 V	off
Voltage 2	1 V	off
Period		off
Delay time	15n - t_delay s	off
Rise time	50p s	off
Fall time	50p s	off
Pulse width	100n s	off
Temperature coefficient 1		off
Temperature coefficient 2		off
Nominal temperature		off
Type of rising & falling edge		off

OK

Cancel

Apply

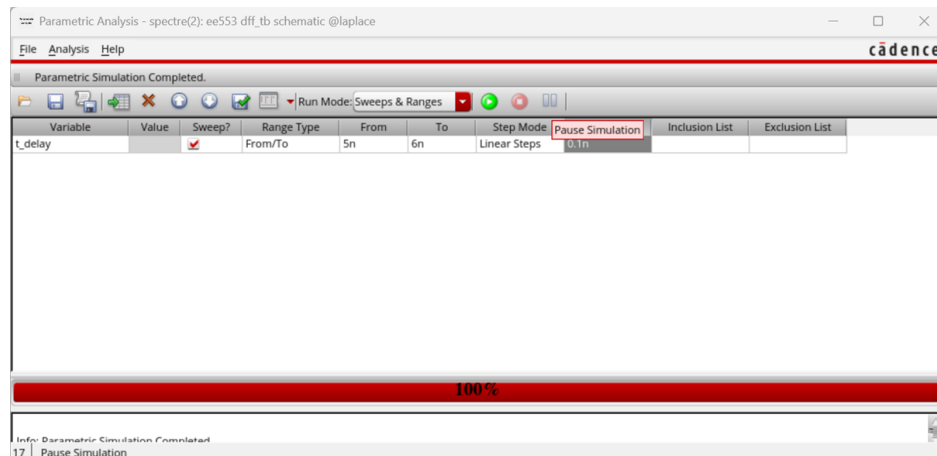
Defaults

Previous

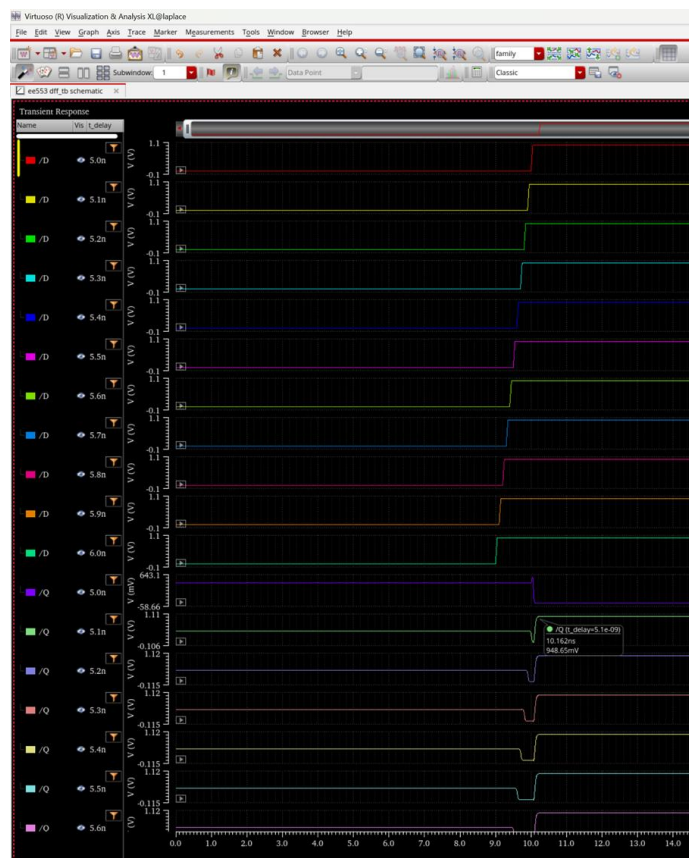
Next

Help

The t_{delay} was swept using a parametric sweep close to the rising edge.



It was discovered that the output Q is triggered when the $t_{\text{delay}}=5.1\text{ns}$:



Now, that we have discovered the delay time needed to trigger the output Q, to find the hold time we set the characteristics of the input D as follows:

CDF Parameter	Value	Display
Frequency name for 1/period		off 
Noise file name		off 
Number of noise/freq pairs	0	off 
DC voltage		off 
AC magnitude		off 
AC phase		off 
XF magnitude		off 
PAC magnitude		off 
PAC phase		off 
Voltage 1	0 V	off 
Voltage 2	1 V	off 
Period		off 
Delay time	15n - 5.1n s	off 
Rise time	50p s	off 
Fall time	50p s	off 
Pulse width	t_hold s	off 
Temperature coefficient 1		off 
Temperature coefficient 2		off 
Nominal temperature		off 
Type of rising & falling edge		off 

Finally, the hold time was discovered to be **7p** seconds.



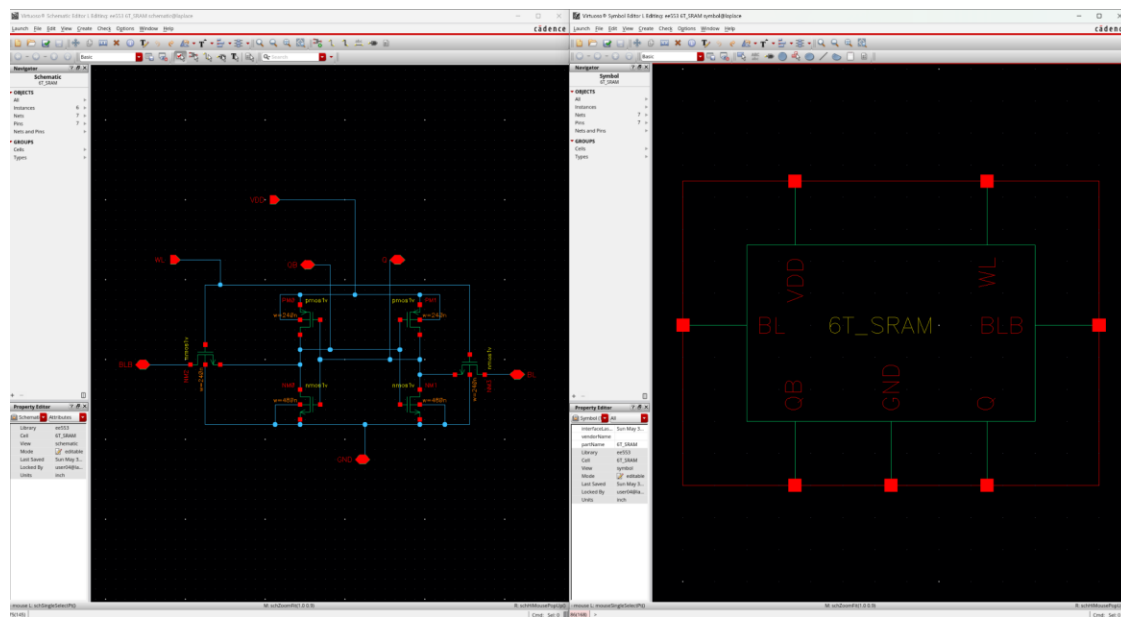
Part III 16 × 16 SRAM

Navigation through the SRAM Files:

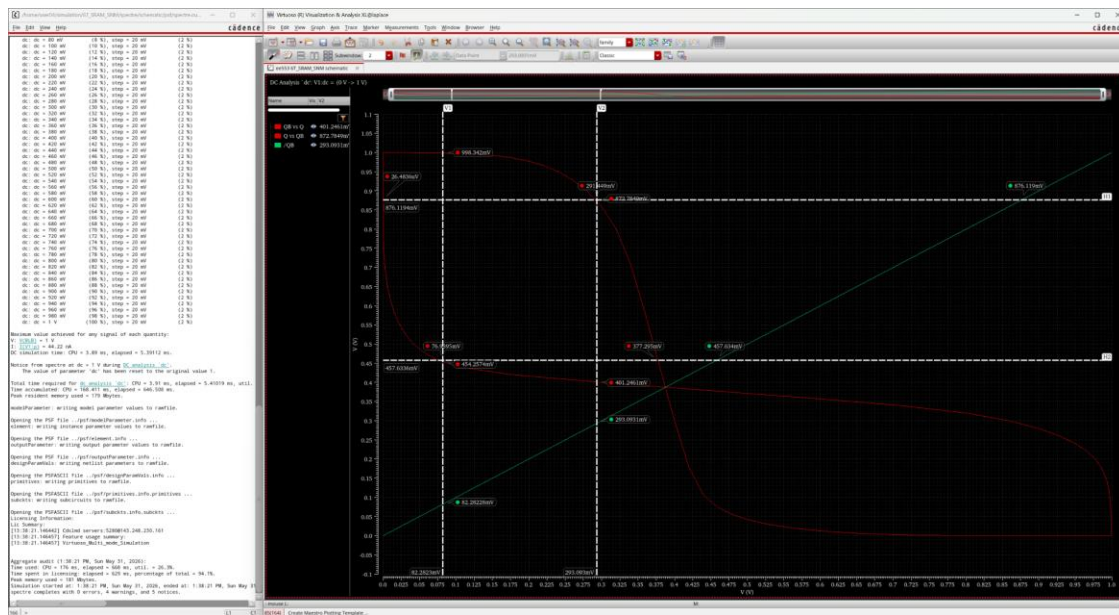
- SRAM (main SRAM chip)
- SRAM_DECODER
- SRAM_NAND5
- SRAM_INV
- SRAM_BLSA
- SRAM_PRECHARGE
- SRAM_WRITE_DRIVER
- 6T_SRAM

1. SRAM Bitcell [Schematic/ADE]

6T SRAM Schematic and Symbol

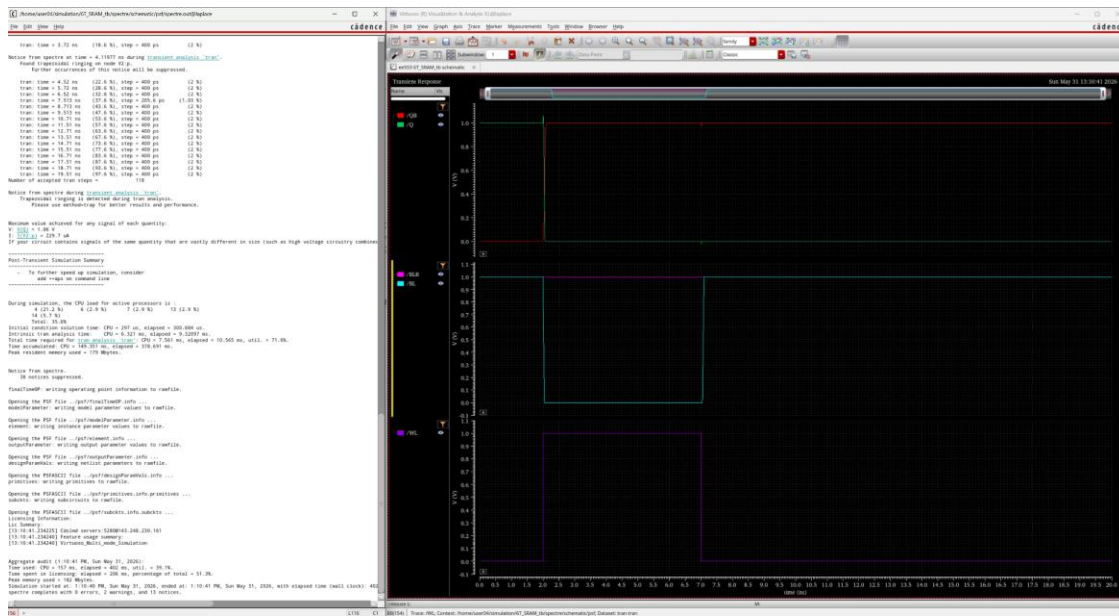


SNM = 210.8107 mV



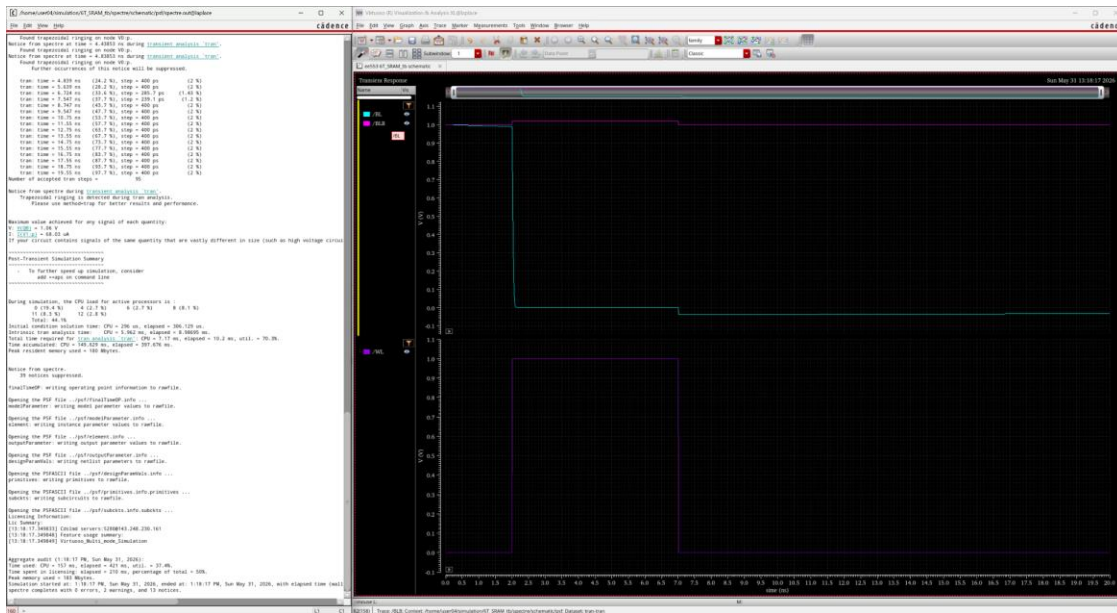
6T SRAM Write 0 Operation

Initial Conditions: Q set to **1V**. QB set **0V**.

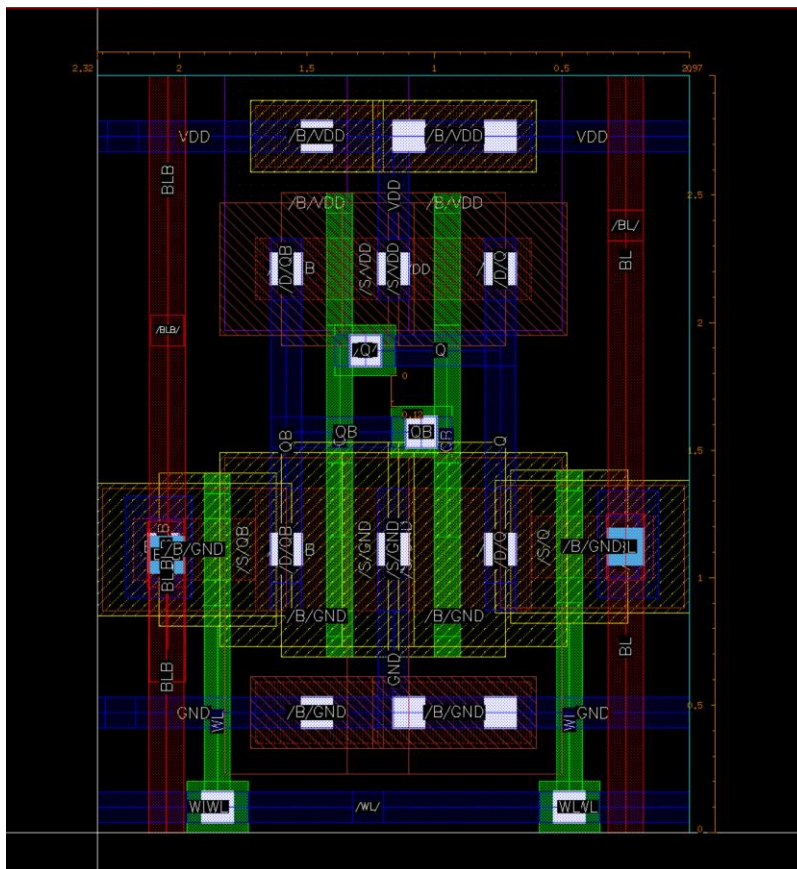


6T SRAM Read Operation

Initial Conditions: BL set to **1V**. BLB set to **1V**. (This mimics the precharge phase). Q set to **0V** and QB to **1V**.



6T SRAM Layout:



DRC and LVS Results

```

/home/user04/gpdk090_v4.6/6T_SRAM.log@laplace
File Edit View Help
cadence

( 822) BONDPAD.SP.6: Bondpad Metal7 to Bondpad      0      0
( 823) BONDPAD.SP.6: Bondpad Metal7 to Bondpad      0      0
( 824) BONDPAD.SP.6: Bondpad Metal8 to Bondpad      0      0
( 825) BONDPAD.SP.6: Bondpad Metal8 to Bondpad      0      0
( 826) BONDPAD.SP.7: Bondpad Via1 array to Bond      0      0
( 827) BONDPAD.SP.7: Bondpad Via2 array to Bond      0      0
( 828) BONDPAD.SP.7: Bondpad Via3 array to Bond      0      0
( 829) BONDPAD.SP.7: Bondpad Via4 array to Bond      0      0
( 830) BONDPAD.SP.7: Bondpad Via5 array to Bond      0      0
( 831) BONDPAD.SP.7: Bondpad Via6 array to Bond      0      0
( 832) BONDPAD.SP.7: Bondpad Via7 array to Bond      0      0
( 833) BONDPAD.SP.7: Bondpad Via8 array to Bond      0      0
( 834) OXIDER.W.1.1: Oxide resistor width must     0      0
( 835) POLYR.W.1.1: Poly resistor width must be    0      0
-----
Total errors:                                0      0
-----

Finished creating Error Database ...

Writing Report into ./6T_SRAM.err ...

avrpt cpu sec:   0.18 elapsed:    0 virtual:   93M

Finished /eda/CADENCE/ASSURA//tools.lnx86/assura/bin/avrpt

Starting /eda/CADENCE/ASSURA//tools.lnx86/assura/bin/avcallproc /home/user04/gpdk090_v4.6/6T_SRAM.rsf -restart -cdslib /home/user04/g
2026/05/31 14:09:54 WARNING This OS does not appear to be a Cadence supported Linux configuration.
2026/05/31 14:09:54 For more info, please run CheckSysConf in <cdsRoot/tools.lnx86/bin/checkSysConf <productId>

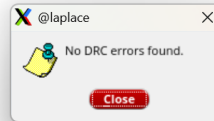
Loading tech rule set file : /home/user04/gpdk090_v4.6/assura/techRuleSets
Loading tech rule set file : /home/user04/gpdk090_v4.6/assura/techRuleSets
Loading tech rule set file : /home/user04/gpdk090_v4.6/assura/techRuleSets
Finished /eda/CADENCE/ASSURA//tools.lnx86/assura/bin/avcallproc

Run ended: Sun May 31 14:09:54 2026

***** Assura terminated normally *****

```

174 | L5,436 C1



```

/home/user04/gpdk090_v4.6/6T_SRAM.log@laplace
File Edit View Help
cadence

*WARNING* envSetVal: could not find tool[.partition] 'spectre.env0pts'
*WARNING* Cannot find /eda/CADENCE/ASSURA//tools.lnx86/dfII/etc/tools/hspiceD directory to load environment variables
Loading gpdk090/gpdk090_PDKRegistrations.il ... done!

*****
*          Cadence Design Systems, Inc.          *
*          Generic 90nm PDK                      *
*          (gpdk090)                             *
*****

Version : 4.6

Build date: 10-FEB-2011

done!
Loaded gpdk090/libInit.il successfully!

Finished /eda/CADENCE/ASSURA//tools.lnx86/assura/bin/ercChk

Starting /eda/CADENCE/ASSURA//tools.lnx86/assura/bin/avcallproc /
2026/05/31 14:10:29 WARNING This OS does not appear to be a Cadence supported Linux configuration.
2026/05/31 14:10:29 For more info, please run CheckSysConf in <cdsRoot/tools.lnx86/bin/checkSysConf <productId>

Loading tech rule set file : /home/user04/gpdk090_v4.6/assura/tec
Loading tech rule set file : /home/user04/gpdk090_v4.6/assura/tec
Loading tech rule set file : /home/user04/gpdk090_v4.6/assura/tec
Finished /eda/CADENCE/ASSURA//tools.lnx86/assura/bin/avcallproc

Assura LVS terminated normally.

Run ended: Sun May 31 14:10:30 2026

***** Assura terminated normally *****

```

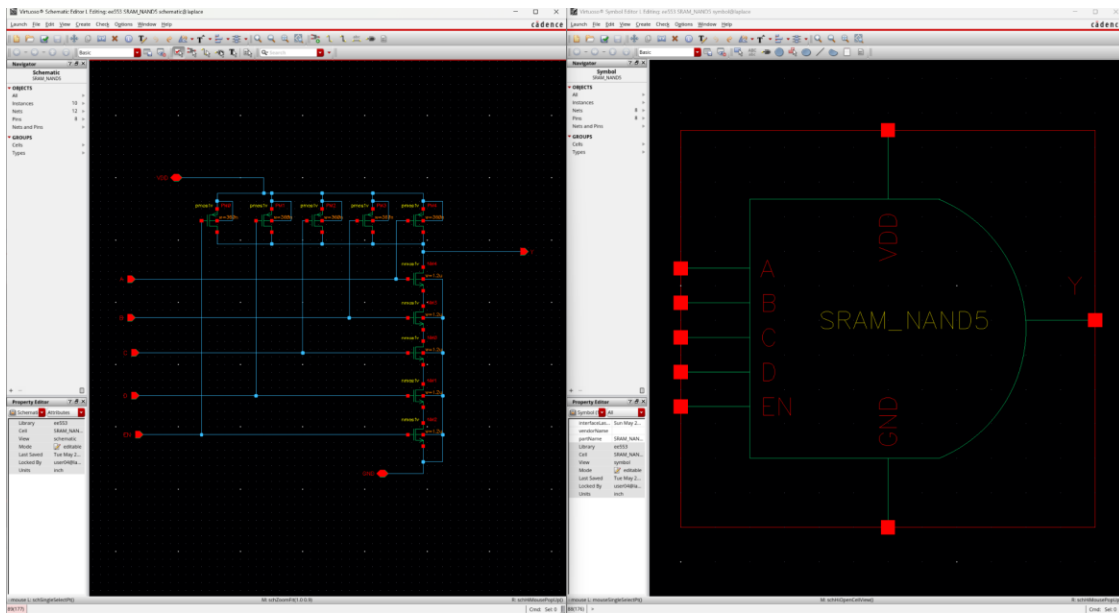
175 | L2,126 C1

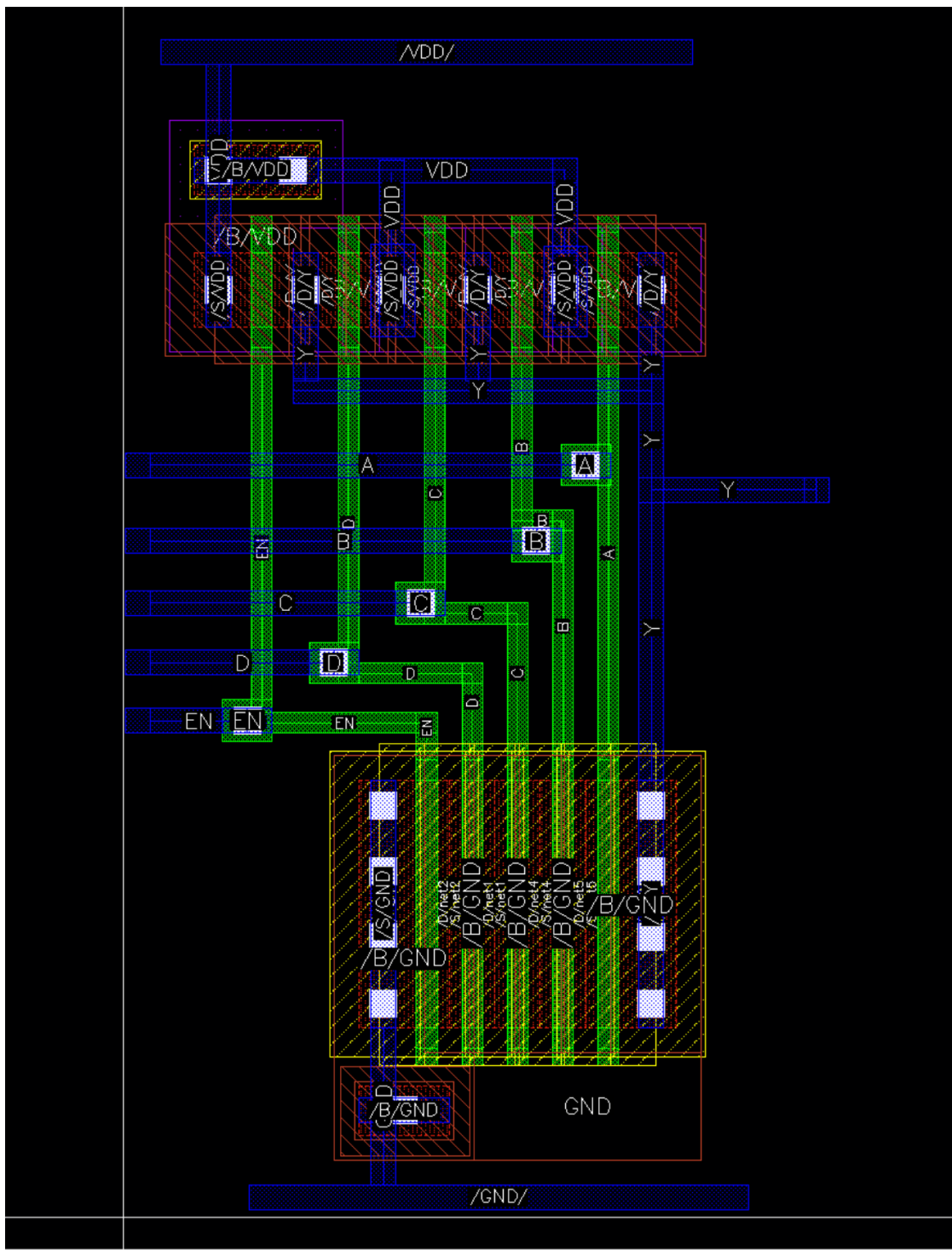


2. SRAM Column Design [Schematic/ADE]

4-to-16 Decoder with enable signal

For the decoder with the enable signal I have build the 5 input NAND gate whose schematic, symbol and layout are shown below.

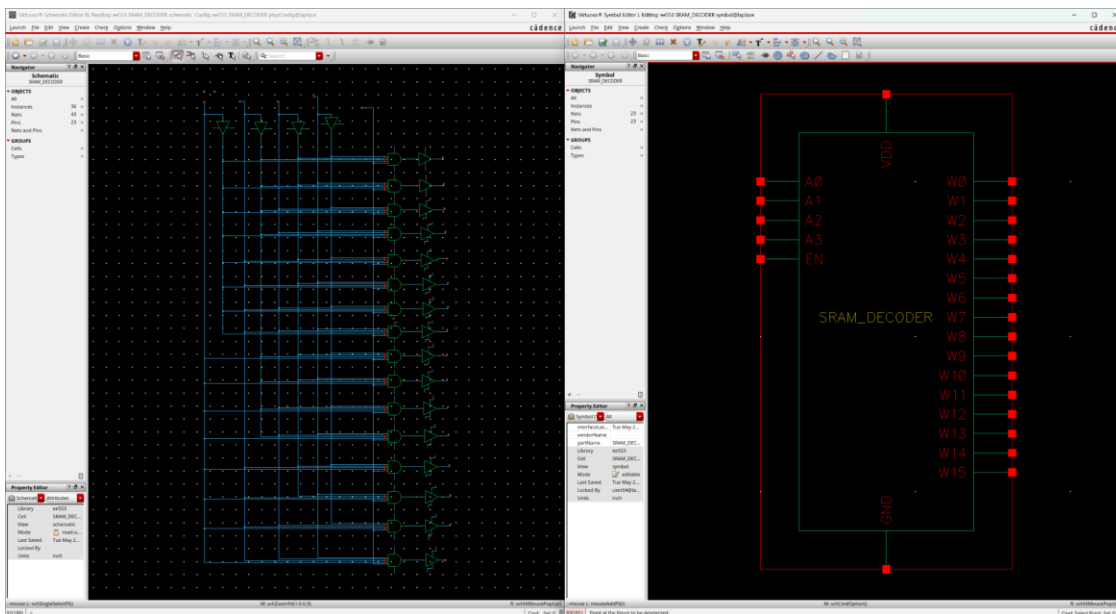




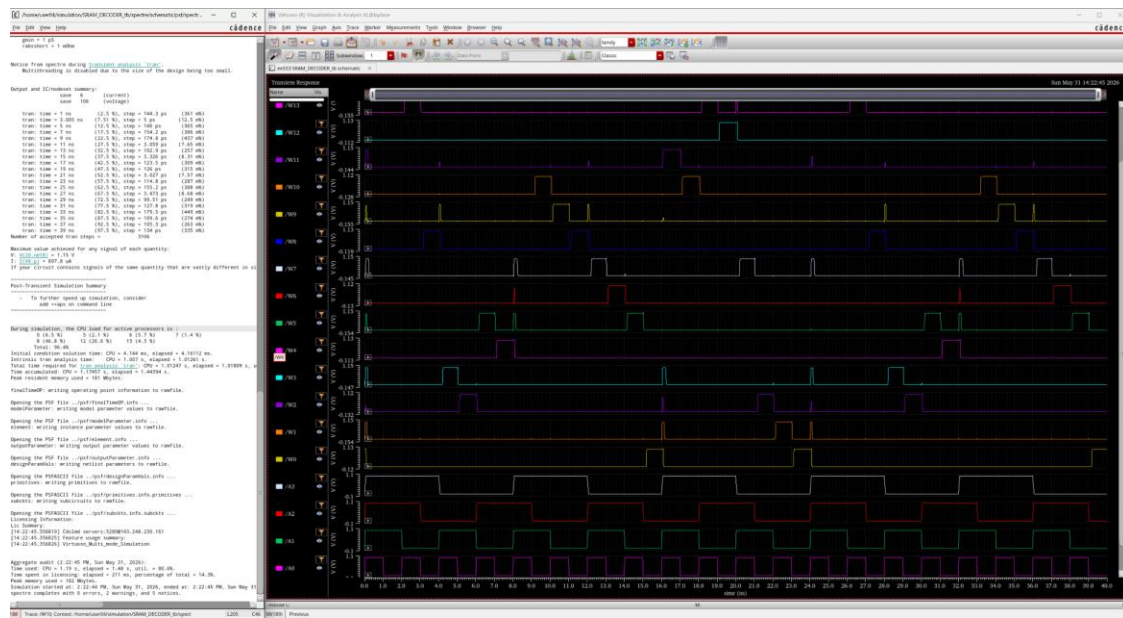
The 4-to-16 decoder truth table:

Enable	Inputs				outputs															
E	I ₃	I ₂	I ₁	I ₀	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
0	X	X	X	X	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
1	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
1	0	1	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
1	0	1	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
1	0	1	1	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
1	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
1	1	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
1	1	0	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
1	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

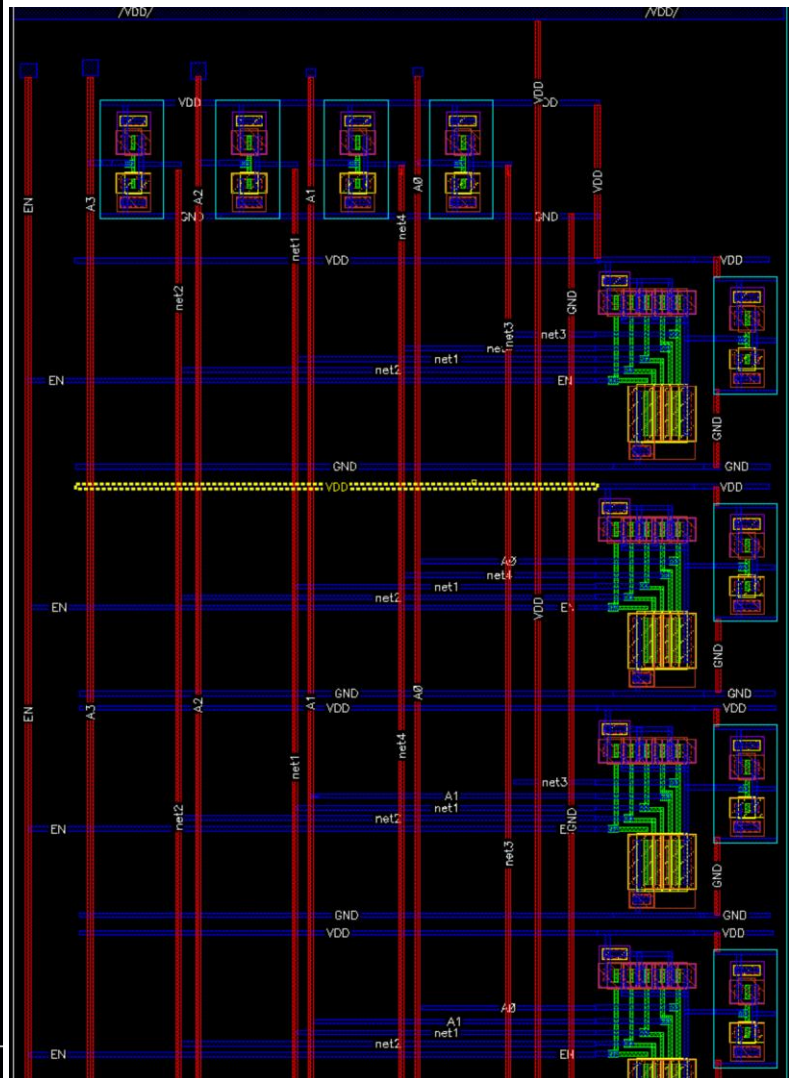
Based on the truth table shown above, the 4-to-16 decoder circuit was built using inverters and NAND5 gates:



Functionality of the decoder is verified as shown in the below transient response while enable signal is on.



4-to-16 Decoder Layout (full and zoomed in)



DRC and LVS Results

```
/home/user04/gpdk090_v4.6/SRAM_DECODER.log@laplace
File Edit View Help
cadence

( 822) BONDPAD.SP.6: Bondpad Meta17 to Bondpad      0      0
( 823) BONDPAD.SP.6: Bondpad Meta17 to Bondpad      0      0
( 824) BONDPAD.SP.6: Bondpad Meta18 to Bondpad      0      0
( 825) BONDPAD.SP.6: Bondpad Meta18 to Bondpad      0      0
( 826) BONDPAD.SP.7: Bondpad Via1 array to Bond      0      0
( 827) BONDPAD.SP.7: Bondpad Via2 array to Bond      0      0
( 828) BONDPAD.SP.7: Bondpad Via3 array to Bond      0      0
( 829) BONDPAD.SP.7: Bondpad Via4 array to Bond      0      0
( 830) BONDPAD.SP.7: Bondpad Via5 array to Bond      0      0
( 831) BONDPAD.SP.7: Bondpad Via6 array to Bond      0      0
( 832) BONDPAD.SP.7: Bondpad Via7 array to Bond      0      0
( 833) BONDPAD.SP.7: Bondpad Via8 array to Bond      0      0
( 834) OXIDER.W.1.1: Oxide resistor width must be  0      0
( 835) POLYR.W.1.1: Poly resistor width must be    0      0
-----
Total errors:                                0      0

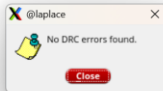
Finished creating Error Database ...
Writing Report into ./SRAM_DECODER.err ...

avrpt cpu sec: 0.19 elapsed: 1 virtual: 93M
Finished /eda/CADENCE/ASSURA/tools.lnx86/assura/bin/avrpt

Starting /eda/CADENCE/ASSURA/tools.lnx86/assura/bin/avcallproc /home/user04/gpdk090_v4.6/SRAM_DECODER.rsf -restart -cdslib /home/user04/gpdk090_v4.6/cdslib
2026/05/31 14:39:19 WARNING This OS does not appear to be a Cadence supported Linux configuration.
2026/05/31 14:39:19 For more info, please run CheckSysConf in <cdsRoot/tools.lnx86/bin/checkSysConf <productId>

Loading tech rule set file : /home/user04/gpdk090_v4.6/assura/techRuleSets
Loading tech rule set file : /home/user04/gpdk090_v4.6/assura/techRuleSets
Loading tech rule set file : /home/user04/gpdk090_v4.6/assura/techRuleSets
Finished /eda/CADENCE/ASSURA/tools.lnx86/assura/bin/avcallproc

Run ended: Sun May 31 14:39:19 2026
***** Assura terminated normally *****
```



```
/home/user04/gpdk090_v4.6/SRAM_DECODER.log@laplace
File Edit View Help
cadence

***WARNING*** envSetVal: could not find tool[.partition] 'spectre.env0pts'
***WARNING*** Cannot find /eda/CADENCE/ASSURA/tools.lnx86/dft/retc/tools/hspiceD directory to load environment variables
Loading gpdk090/gpdk090_PDKRegistrations.il ... done!

*****
* Cadence Design Systems, Inc.
*
* Generic 90nm PDK
* (gpdk090)
*****

Version : 4.6
Build date: 10-FEB-2011

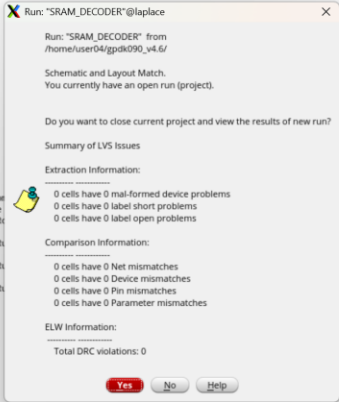
done!
Loaded gpdk090/libInit.il successfully!
Finished /eda/CADENCE/ASSURA/tools.lnx86/assura/bin/ercChk

Starting /eda/CADENCE/ASSURA/tools.lnx86/assura/bin/avcallproc /home/user04/gpdk090_v4.6/SRAM_DECODER.rsf -restart -cdslib /home/user04/gpdk090_v4.6/cdslib
2026/05/31 14:40:16 WARNING This OS does not appear to be a Cadence supported Linux configuration.
2026/05/31 14:40:16 For more info, please run CheckSysConf in <cdsRoot/tools.lnx86/bin/checkSysConf <productId>

Loading tech rule set file : /home/user04/gpdk090_v4.6/assura/techRuleSets
Loading tech rule set file : /home/user04/gpdk090_v4.6/assura/techRuleSets
Loading tech rule set file : /home/user04/gpdk090_v4.6/assura/techRuleSets
Finished /eda/CADENCE/ASSURA/tools.lnx86/assura/bin/avcallproc

Assura LVS terminated normally.

Run ended: Sun May 31 14:40:17 2026
***** Assura terminated normally *****
```

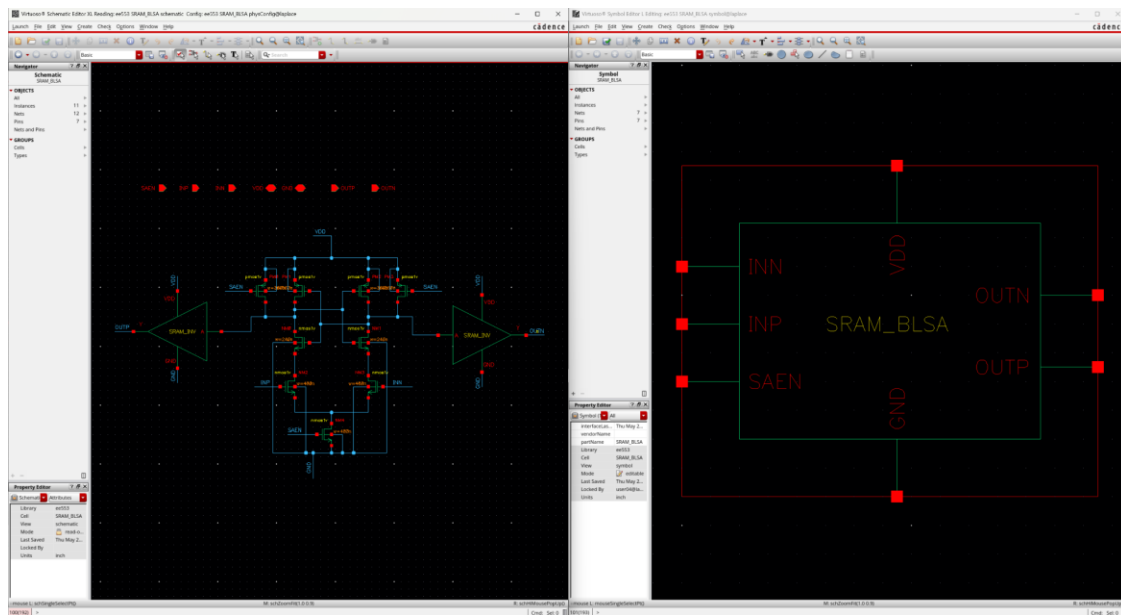


BLSA (Bit Line Sense Amplifier)

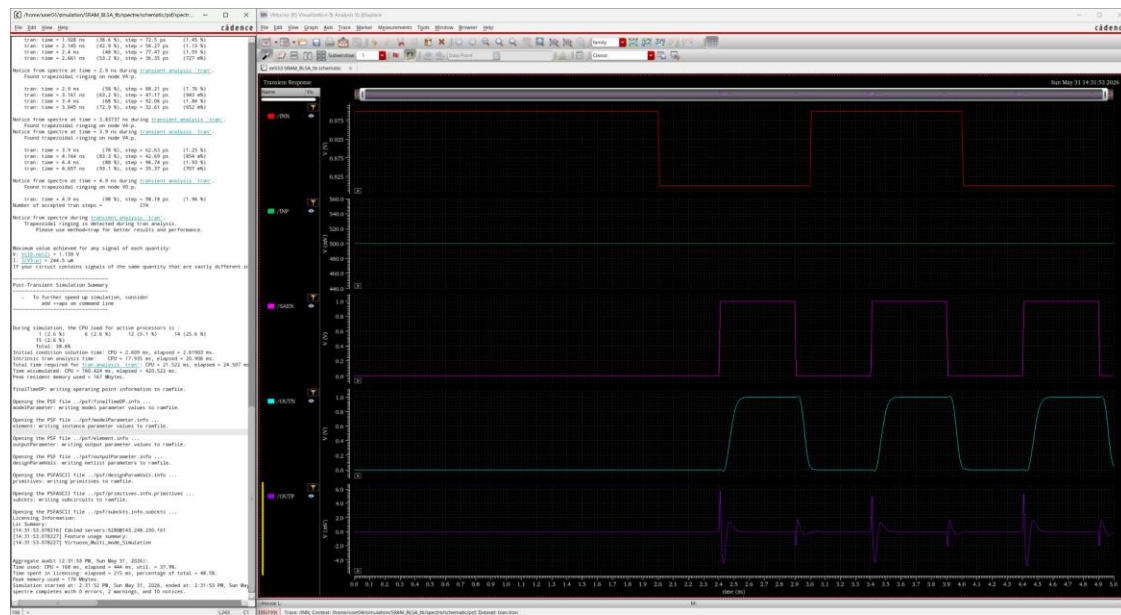
How the BLSA Works

The sense amplifier operates in two phases. During the **precharge phase (SAEN = 0 V)**, the circuit remains inactive because the bottom NMOS transistor is turned off, preventing any current flow through the amplifier. At the same time, the precharge PMOS transistors are turned on, pulling both internal nodes up to VDD and resetting the circuit to a balanced state. As a result, the output inverters drive both OUTP and OUTN to 0 V. During the **evaluation phase (SAEN = VDD)**, the precharge PMOS transistors turn off while the bottom NMOS transistor turns on, enabling current flow toward ground. The discharge current can flow through either the left branch, controlled by INP, or the right branch, controlled by INN (RBL). With INP fixed at 0.5 VDD and INN reduced to 0.4 VDD, the left NMOS conducts more strongly, causing the left internal node to discharge faster than the right. This small voltage difference is rapidly amplified by the cross-coupled inverter pair through positive feedback, forcing the left node to ground and the right node to VDD. The output inverters then buffer these voltages, producing stable complementary digital outputs at OUTP and OUTN.

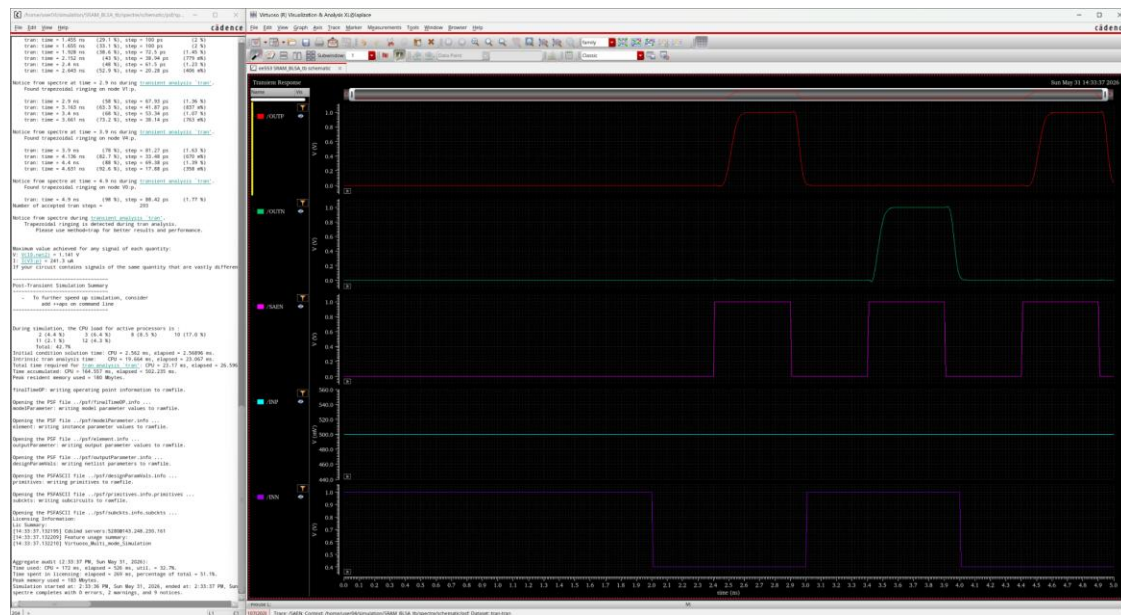
BLSA Schematic and Symbol



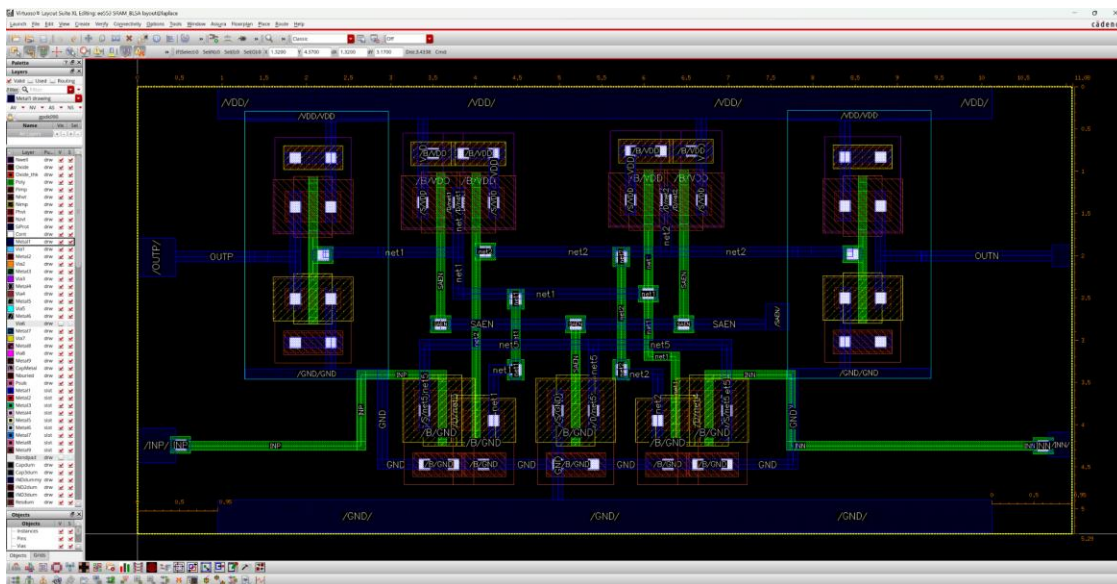
INP < INN (0.5V < 0.8V)



INP > INN (0.5V > 0.4V)



BLSA Layout



DRC and LVS Results

```

/home/user04/gpdk090_v4.6/SRAM_BLSA.log@laplace
File Edit View Help
cadence

( 822) BONDPAD.SP.6: Bondpad Metal7 to Bondpad      0      0
( 823) BONDPAD.SP.6: Bondpad Metal7 to Bondpad      0      0
( 824) BONDPAD.SP.6: Bondpad Metal8 to Bondpad      0      0
( 825) BONDPAD.SP.6: Bondpad Metal8 to Bondpad      0      0
( 826) BONDPAD.SP.7: Bondpad Via1 array to Bond      0      0
( 827) BONDPAD.SP.7: Bondpad Via2 array to Bond      0      0
( 828) BONDPAD.SP.7: Bondpad Via3 array to Bond      0      0
( 829) BONDPAD.SP.7: Bondpad Via4 array to Bond      0      0
( 830) BONDPAD.SP.7: Bondpad Via5 array to Bond      0      0
( 831) BONDPAD.SP.7: Bondpad Via6 array to Bond      0      0
( 832) BONDPAD.SP.7: Bondpad Via7 array to Bond      0      0
( 833) BONDPAD.SP.7: Bondpad Via8 array to Bond      0      0
( 834) OXIDER.W.1.1: Oxide resistor width must      0      0
( 835) POLYR.W.1.1: Poly resistor width must be     0      0
-----
Total errors:                                0      0
-----

Finished creating Error Database ...

Writing Report into ./SRAM_BLSA.err ...

avrpt cpu sec:  0.18 elapsed:  1 virtual:  93M

Finished /eda/CADENCE/ASSURA//tools.lnx86/assura/bin/avrpt

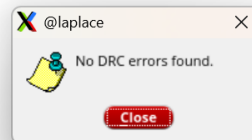
Starting /eda/CADENCE/ASSURA//tools.lnx86/assura/bin/avcallproc /home/user04/gpdk090_v4.6/SRAM_BLSA.rsf -restart -cdslib /home/user04
2026/05/31 14:37:02 WARNING This OS does not appear to be a Cadence supported Linux configuration.
2026/05/31 14:37:02 For more info, please run CheckSysConf in <cdsRoot/tools.lnx86/bin/checkSysConf <productId>

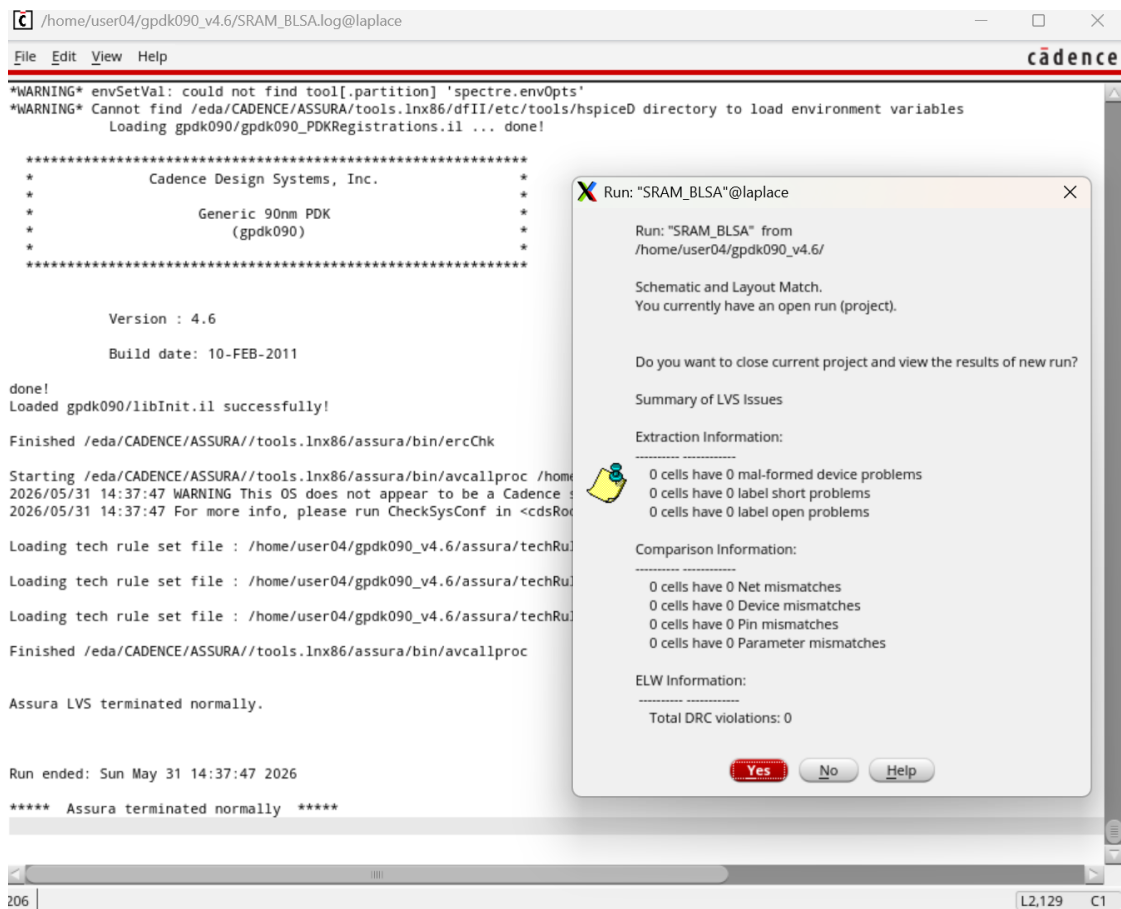
Loading tech rule set file : /home/user04/gpdk090_v4.6/assura/techRuleSets
Loading tech rule set file : /home/user04/gpdk090_v4.6/assura/techRuleSets
Loading tech rule set file : /home/user04/gpdk090_v4.6/assura/techRuleSets
Finished /eda/CADENCE/ASSURA//tools.lnx86/assura/bin/avcallproc

Run ended: Sun May 31 14:37:02 2026

***** Assura terminated normally *****

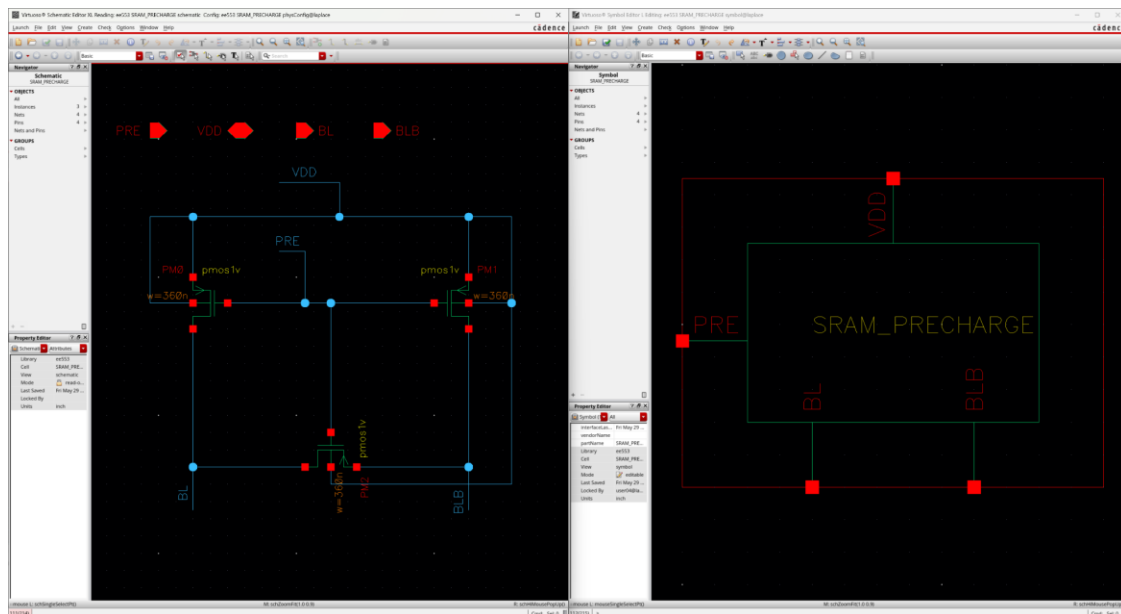
```



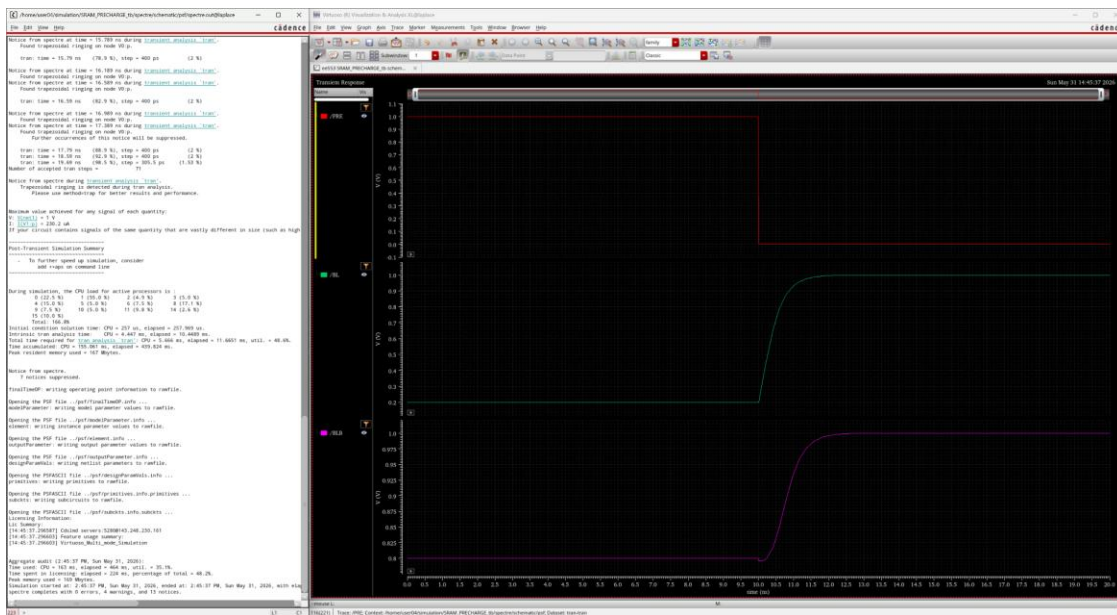


Precharge

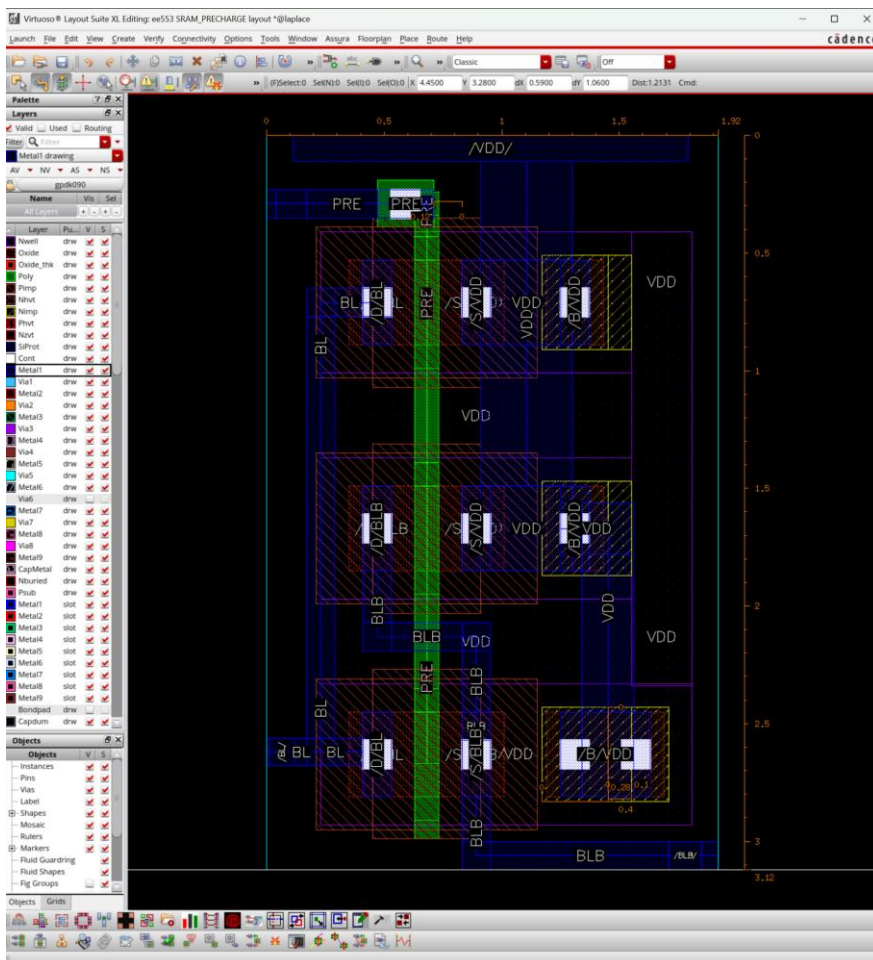
Schematic and Symbol



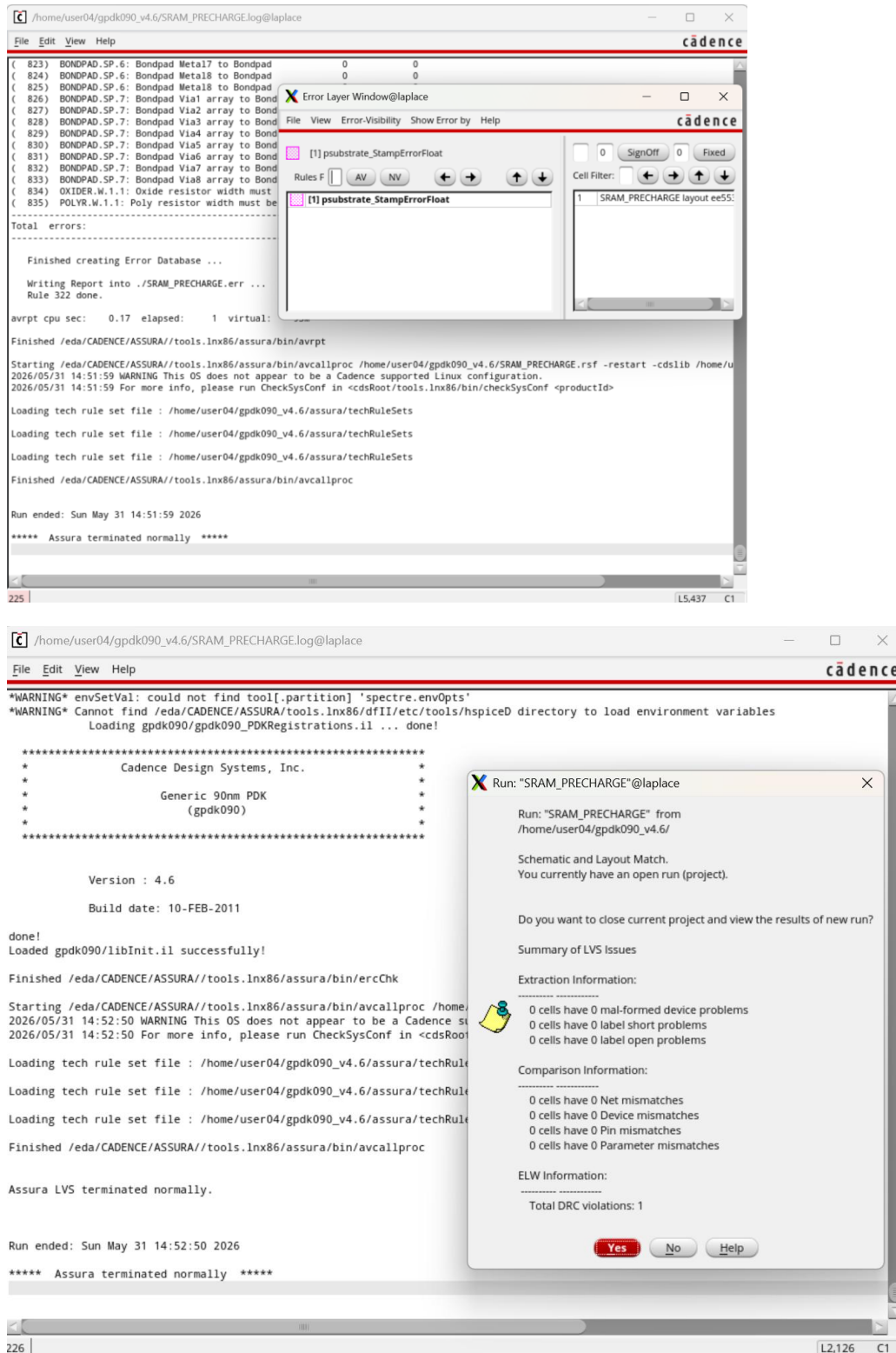
Initial Conditions: BL set to **0.2V**, BLB set to **0.8V**.



Precharge Layout



DRC and LVS Results

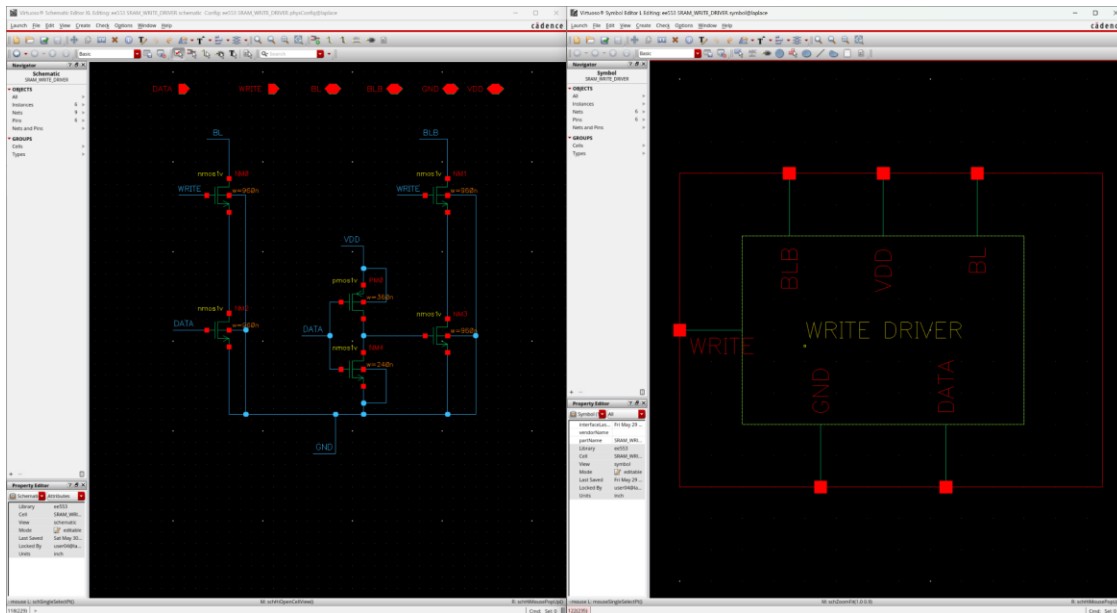


The DRC shows **psubstrate_StampErrorFloat error** this error occurs because we don't have any ground connection in the circuit.

This error further disappears in the full SRAM Layout

Write Driver

Schematic and Symbol

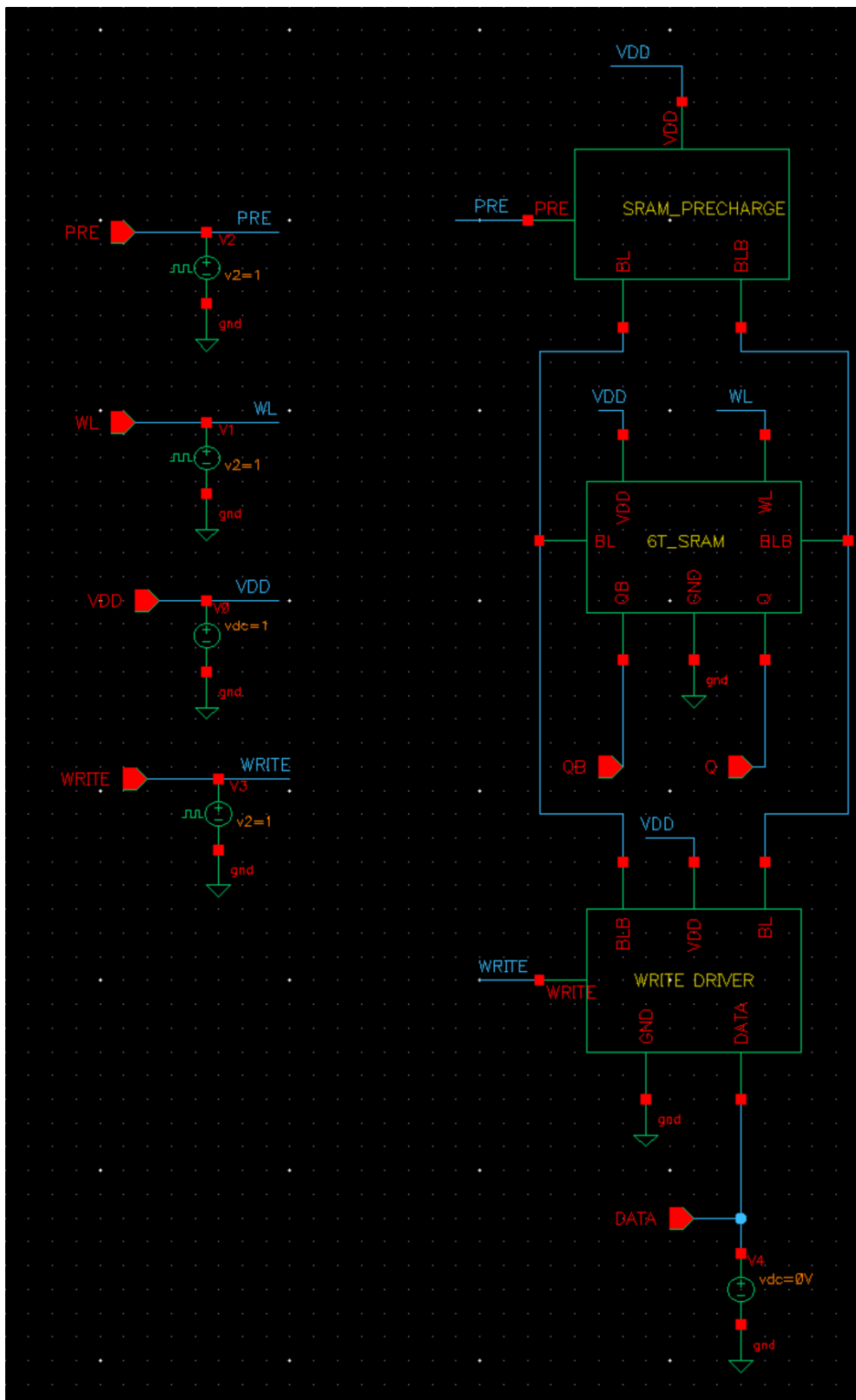


Transient Response

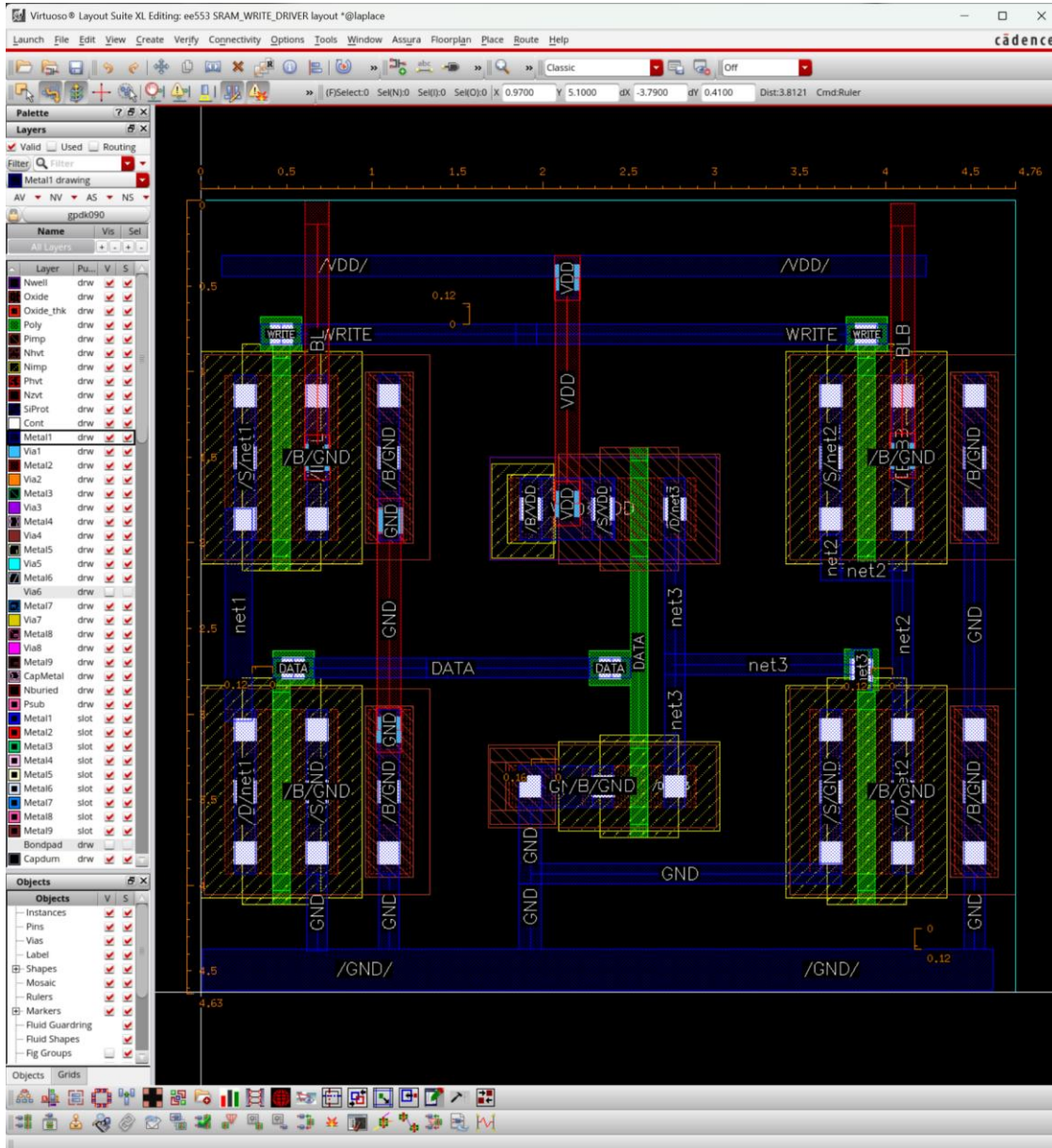
Process:

Here is the exact timing sequence you want to create:

1. Phase 1: Precharge (0ns to 5ns)
 - PRE = 0V (Active).
 - WL = 0V (SRAM cell is disconnected).
 - WRITE = 0V (Write driver is off).
 - *Result:* BL and BLB are charged to 1V.
2. Phase 2: Float (5ns to 6ns)
 - PRE = 1V (Turn off precharge so it doesn't fight the write driver).
 - *Result:* BL and BLB hold their 1V charge.
3. Phase 3: The Write (6ns to 12ns)
 - WL = 1V (Open the gates to the SRAM cell).
 - WRITE = 1V (Turn on your write driver).
 - DATA = 0V (To write a '0').
 - *Result:* Your write driver pulls BL down to 0V. This completely overpowers the tiny transistors inside the SRAM cell, forcing Q to drop to 0V and QB to flip to 1V.
4. Phase 4: Hold (12ns onwards)
 - WL = 0V and WRITE = 0V.
 - *Result:* The cell securely holds its newly written data.



Write Driver Layout



DRC and LVS Results

```
/home/user04/gpd090_v4.6/SRAM_WRITE_DRIVER.log@laplace
File Edit View Help cadence

( 822) BONDPAD.SP.6: Bondpad Metal7 to Bondpad      0      0
( 823) BONDPAD.SP.6: Bondpad Metal7 to Bondpad      0      0
( 824) BONDPAD.SP.6: Bondpad Metal8 to Bondpad      0      0
( 825) BONDPAD.SP.6: Bondpad Metal8 to Bondpad      0      0
( 826) BONDPAD.SP.7: Bondpad Via1 array to Bond      0      0
( 827) BONDPAD.SP.7: Bondpad Via2 array to Bond      0      0
( 828) BONDPAD.SP.7: Bondpad Via3 array to Bond      0      0
( 829) BONDPAD.SP.7: Bondpad Via4 array to Bond      0      0
( 830) BONDPAD.SP.7: Bondpad Via5 array to Bond      0      0
( 831) BONDPAD.SP.7: Bondpad Via6 array to Bond      0      0
( 832) BONDPAD.SP.7: Bondpad Via7 array to Bond      0      0
( 833) BONDPAD.SP.7: Bondpad Via8 array to Bond      0      0
( 834) OXIDER.W.1.1: Oxide resistor width must      0      0
( 835) POLYR.W.1.1: Poly resistor width must be     0      0
-----
Total errors:                                0      0
-----

Finished creating Error Database ...

Writing Report into ./SRAM_WRITE_DRIVER.err ...

avrpt cpu sec: 0.18 elapsed: 0 virtual: 93M

Finished /eda/CADENCE/ASSURA//tools.lnx86/assura/bin/avrpt

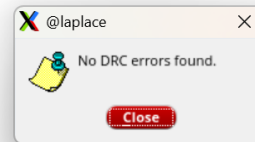
Starting /eda/CADENCE/ASSURA//tools.lnx86/assura/bin/avcallproc /home/user04/gpd090_v4.6/SRAM_WRITE_DRIVER.rs
2026/05/31 15:19:16 WARNING This OS does not appear to be a Cadence supported Linux configuration.
2026/05/31 15:19:16 For more info, please run CheckSysConf in <cdsRoot/tools.lnx86/bin/checkSysConf <productId>

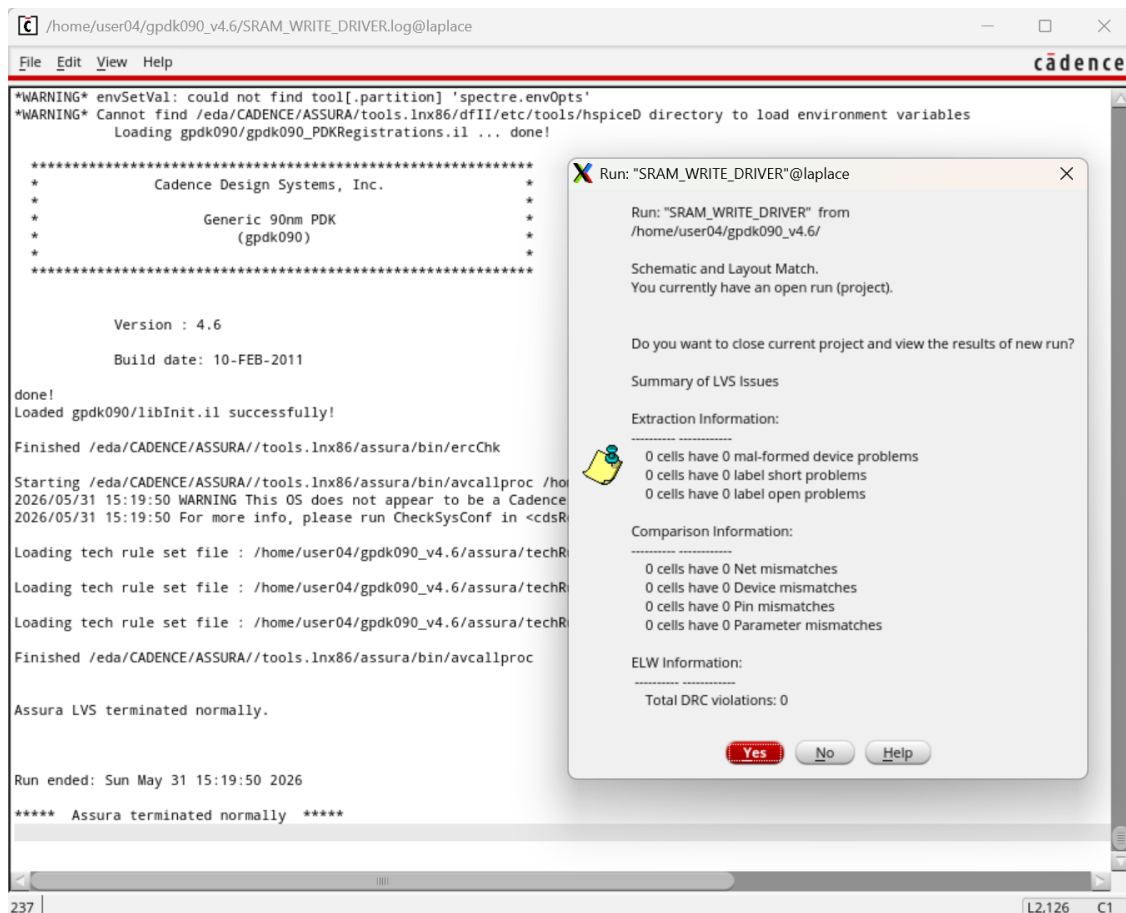
Loading tech rule set file : /home/user04/gpd090_v4.6/assura/techRuleSets
Loading tech rule set file : /home/user04/gpd090_v4.6/assura/techRuleSets
Loading tech rule set file : /home/user04/gpd090_v4.6/assura/techRuleSets
Finished /eda/CADENCE/ASSURA//tools.lnx86/assura/bin/avcallproc

Run ended: Sun May 31 15:19:16 2026

***** Assura terminated normally *****
```

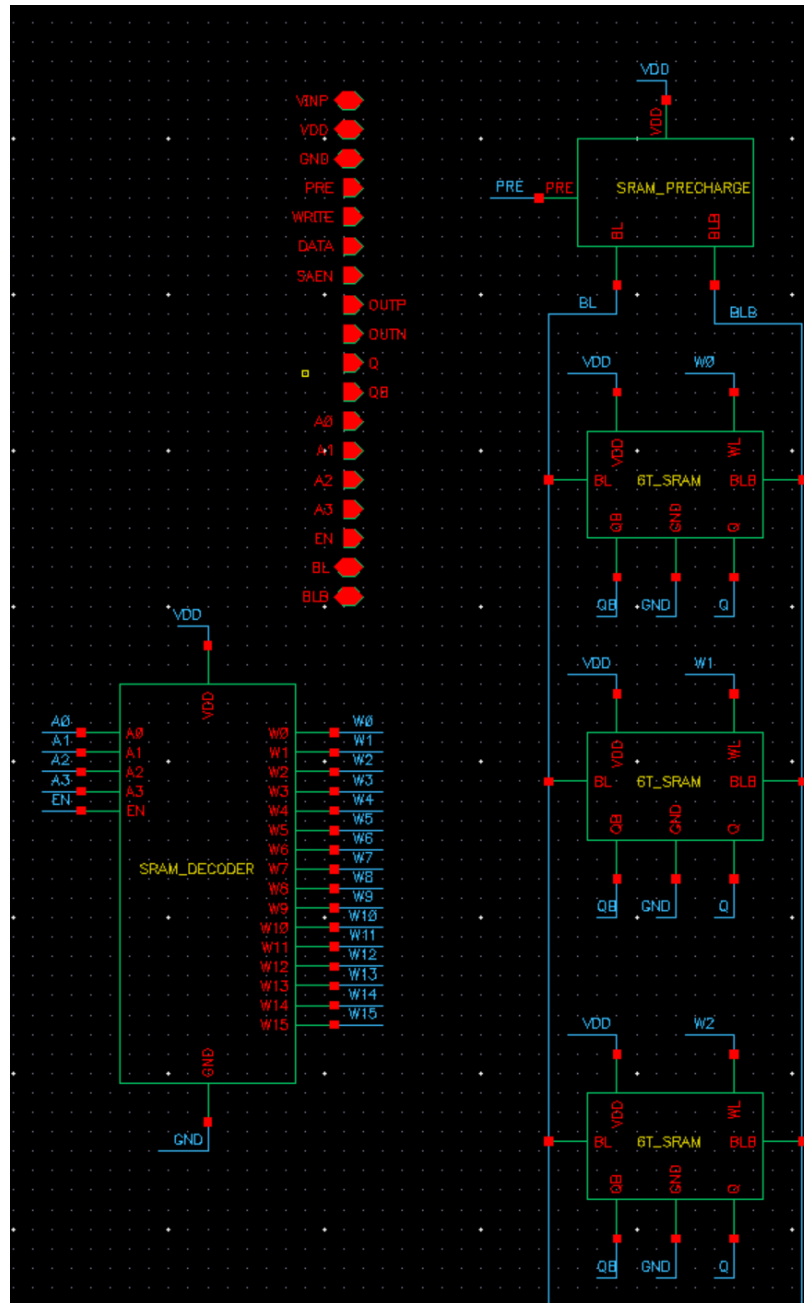
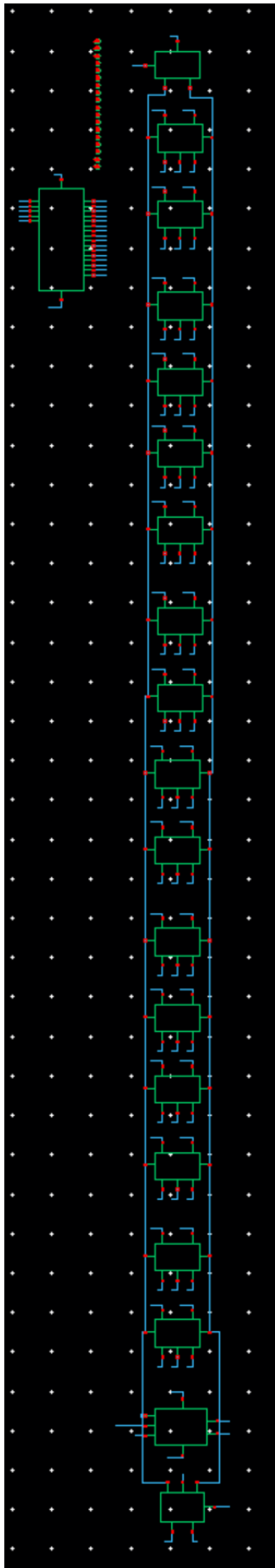
236 | L5,436 C1

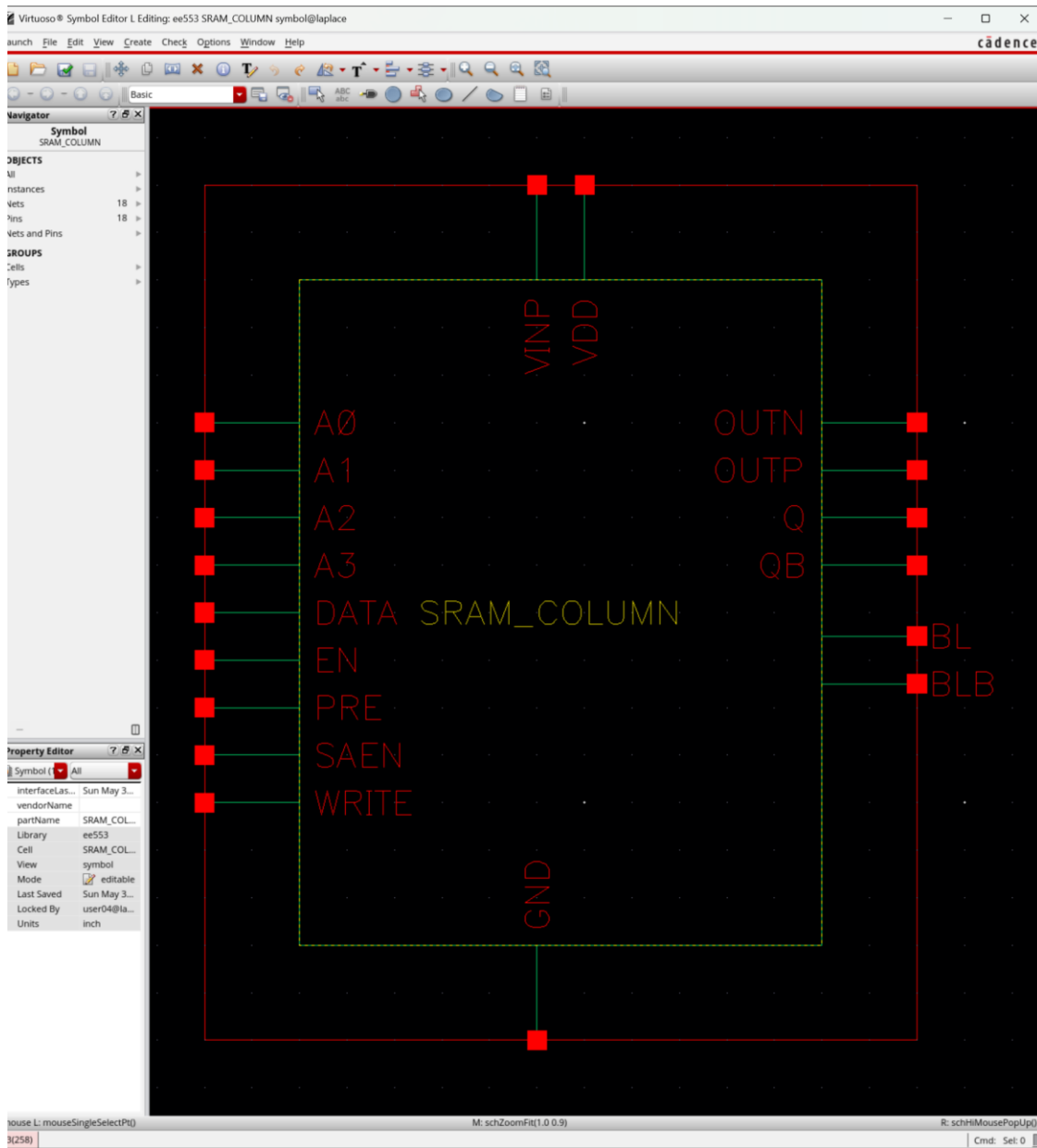




SRAM Column

Schematic and Symbol



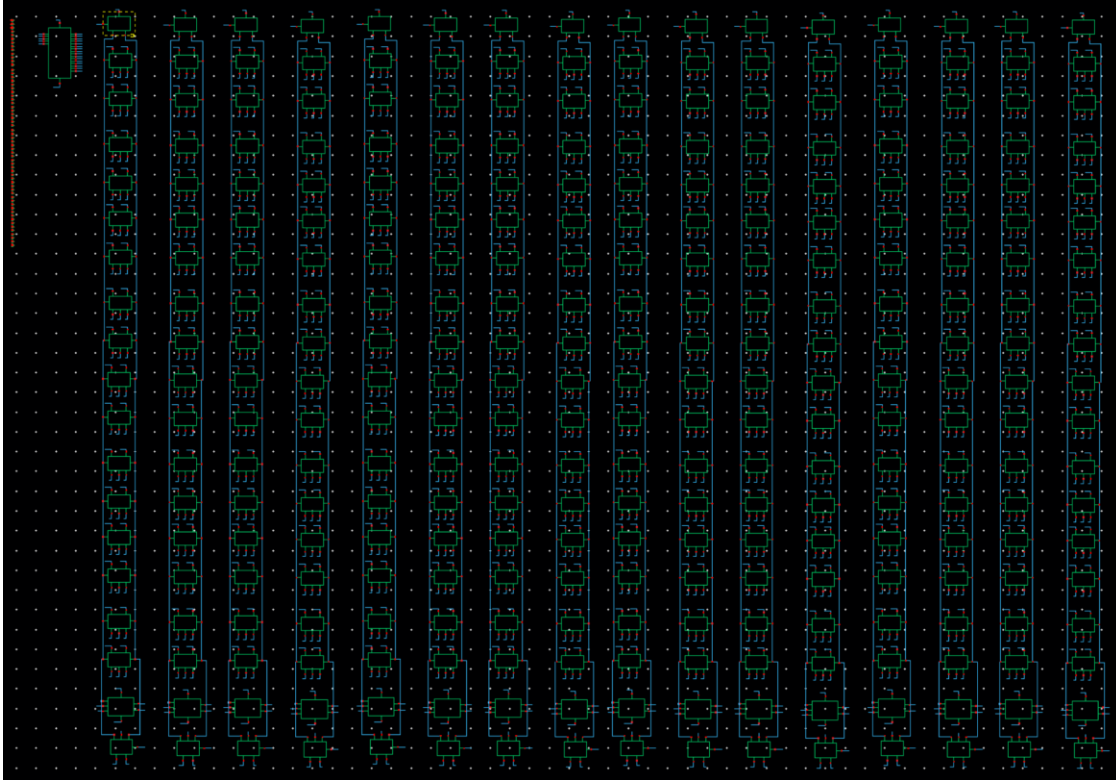


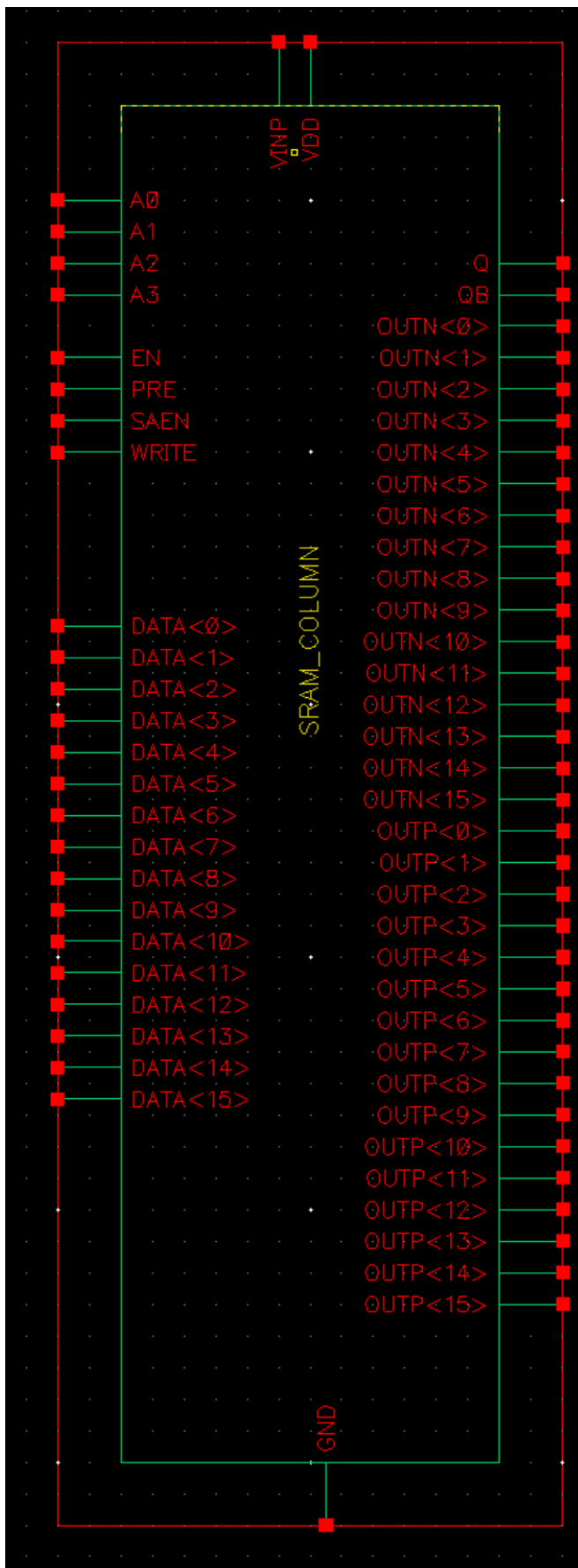
Transient Response



3. 16x16 SRAM array design [Schematic/ADE]

16X16 SRAM Schematic and Symbol





Transient Response (Write and Read WL0 and WL15)



SRAM Layout

